BOP analysis

2026-10-20

```
bop_panel <- readRDS("~/R projects/MD general/bop_panel.rds")
```

```
# making binary variables
bop_panel$Unification_For <- ifelse(bop_panel$Unification == "For", 1, 0)
bop_panel$Nato_For <- ifelse(bop_panel$NATO == "For", 1, 0)
bop_panel$EU_For <- ifelse(bop_panel$EU == "For", 1, 0)
bop_panel$Eurasia_For <- ifelse(bop_panel$Eurasia == "For", 1, 0)
bop_panel$EU_VS <- ifelse(bop_panel$EU_vs_Eurasia == "EU", 1, 0)
bop_panel$Eurasia_VS <- ifelse(bop_panel$EU_vs_Eurasia == "Eurasia", 1, 0)
bop_panel$good <- ifelse(bop_panel$direction == "Good", 1, 0)
bop_panel$bad <- ifelse(bop_panel$direction == "Bad", 1, 0)
```

```
worry_cols <- c("First_Worry", "Second_Worry", "Third_Worry")
bop_panel$poverty <- as.integer(rowSums(bop_panel[worry_cols] == "Poverty", na.rm = TRUE) > 0)
bop_panel$corruption <- as.integer(rowSums(bop_panel[worry_cols] == "Corruption", na.rm = TRUE) > 0)
bop_panel$war <- as.integer(rowSums(bop_panel[worry_cols] == "War in the region", na.rm = TRUE) > 0)
bop_panel$interethnic <- as.integer(rowSums(bop_panel[worry_cols] == "Interethnic relations", na.rm = TR
```

```
bop_panel$Socioeconomic <- fct_relevel(bop_panel$Socioeconomic, "medium")
bop_panel$Education <- fct_relevel(bop_panel$Education, "medium")
```

```
# Coefficient map
coef_model <- c(
  "Age"                           = "Age (per year)",
  "GenderWoman"                   = "Woman",
  "EthnicityRomanian"             = "Romanian",
  "EthnicityRussian"              = "Russian",
  "EthnicityUkrainian"            = "Ukrainian",
  "EthnicityGagauz"               = "Gagauz",
  "EthnicityBulgarian"            = "Bulgarian",
  "Language_QuestionnaireRussian" = "Russian questionnaire",
  "Ecologyrural"                  = "Rural",
  "Educationlow"                = "Education: low",
  "Educationhigh"                 = "Education: high",
  "Socioeconomiclow"            = "Socioeconomic: low",
  "Socioeconomichigh"             = "Socioeconomic: high",
  "Macro_RegionCentre"            = "Centre",
  "Macro_RegionNorth"             = "North",
  "Macro_RegionSouth"             = "South",
  "Macro_RegionGagauzia"          = "Gagauzia",
  "good"            = "Good (Direction)",
  "bad"            = "Bad (Direction)",
  "war"             = "Worry: War",
```

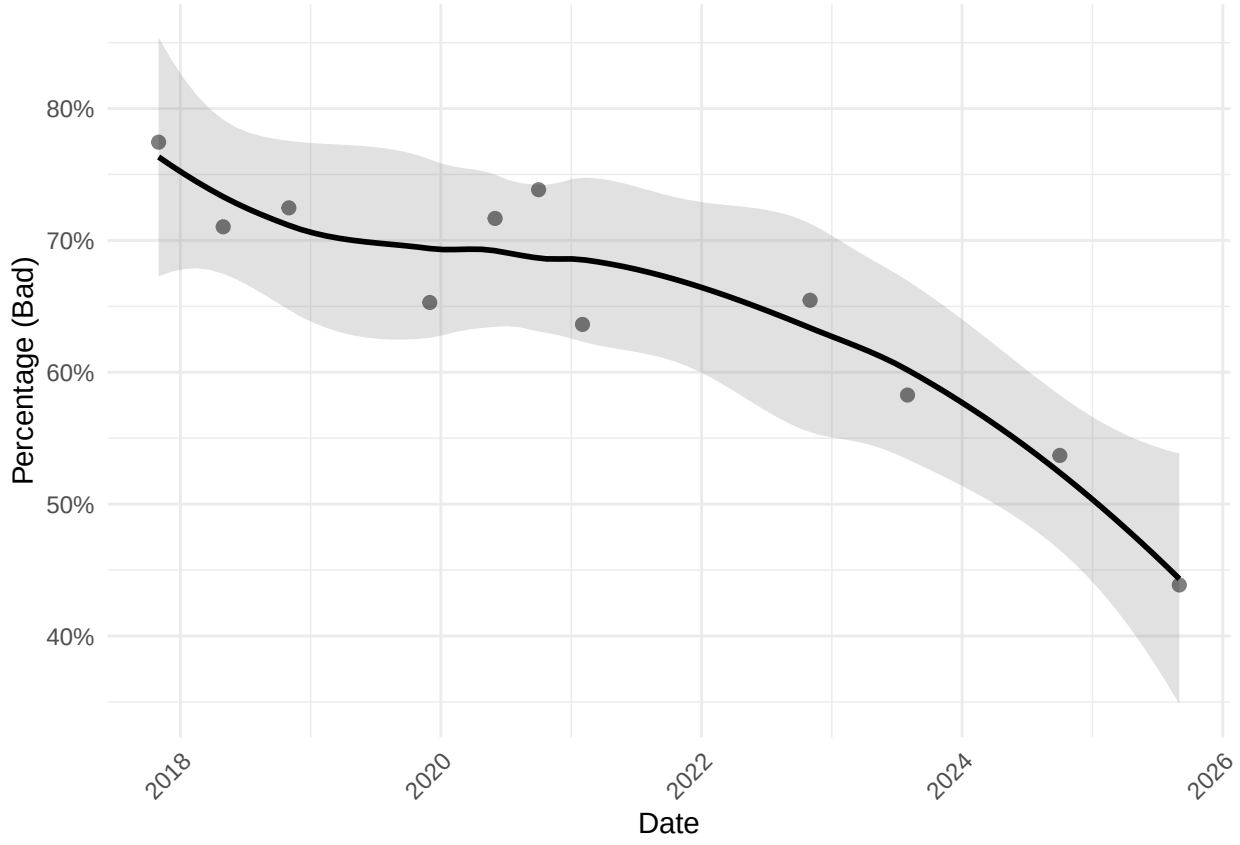
```
  "corruption"      = "Worry: Corruption",
  "interethnic"     = "Worry: Interethnic relations",
  "poverty"         = "Worry: Poverty",
  "Unification_For" = "Unification: in favour",
  "Putin" = "Support: Putin",
  "top_TV"  = "Top media: TV",
  "top_internet" = "Top media: Internet",
  "year"                          = "Year"
)

coef_plot <- c(
  "year"                          = "Year",
  "top_internet"                  = "Top media: Internet",
  "top_TV"                        = "Top media: TV",
  "Putin"                         = "Support: Putin",
  "Unification_For"               = "Unification: in favour",
  "poverty"                       = "Worry: Poverty",
  "interethnic"                   = "Worry: Interethnic relations",
  "corruption"                    = "Worry: Corruption",
  "war"                           = "Worry: War",
  "bad"                           = "Bad (Direction)",
  "good"                          = "Good (Direction)",
  "Macro_RegionGagauzia"          = "Gagauzia",
  "Macro_RegionSouth"             = "South",
  "Macro_RegionNorth"             = "North",
  "Macro_RegionCentre"            = "Centre",
  "Socioeconomichigh"             = "Socioeconomic: high",
  "Socioeconomiclow"              = "Socioeconomic: low",
  "Educationhigh"                 = "Education: high",
  "Educationlow"                  = "Education: low",
  "Ecologyrural"                  = "Rural",
  "Language_QuestionnaireRussian" = "Russian questionnaire",
  "EthnicityBulgarian"            = "Bulgarian",
  "EthnicityGagauz"               = "Gagauz",
  "EthnicityUkrainian"            = "Ukrainian",
  "EthnicityRussian"              = "Russian",
  "EthnicityRomanian"             = "Romanian",
  "GenderWoman"                   = "Woman",
  "Age"                           = "Age (per year)"
)
```

Direction

```
direction_df <- bop_panel %>%
  group_by(wave, date) %>%
  summarise(
    pct_bad = mean(direction == "Bad", na.rm = TRUE),
    pct_good = mean(direction == "Good", na.rm = TRUE),
  )
```

```r
ggplot(direction_df, aes(x = date, y = pct_bad)) +
  geom_smooth(se = TRUE, color = "black", alpha = 0.3) +
  geom_point(size = 2, color = "black", alpha = 0.5) +
  theme_minimal() +
  scale_y_continuous(labels = scales::percent) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage (Bad)")
```



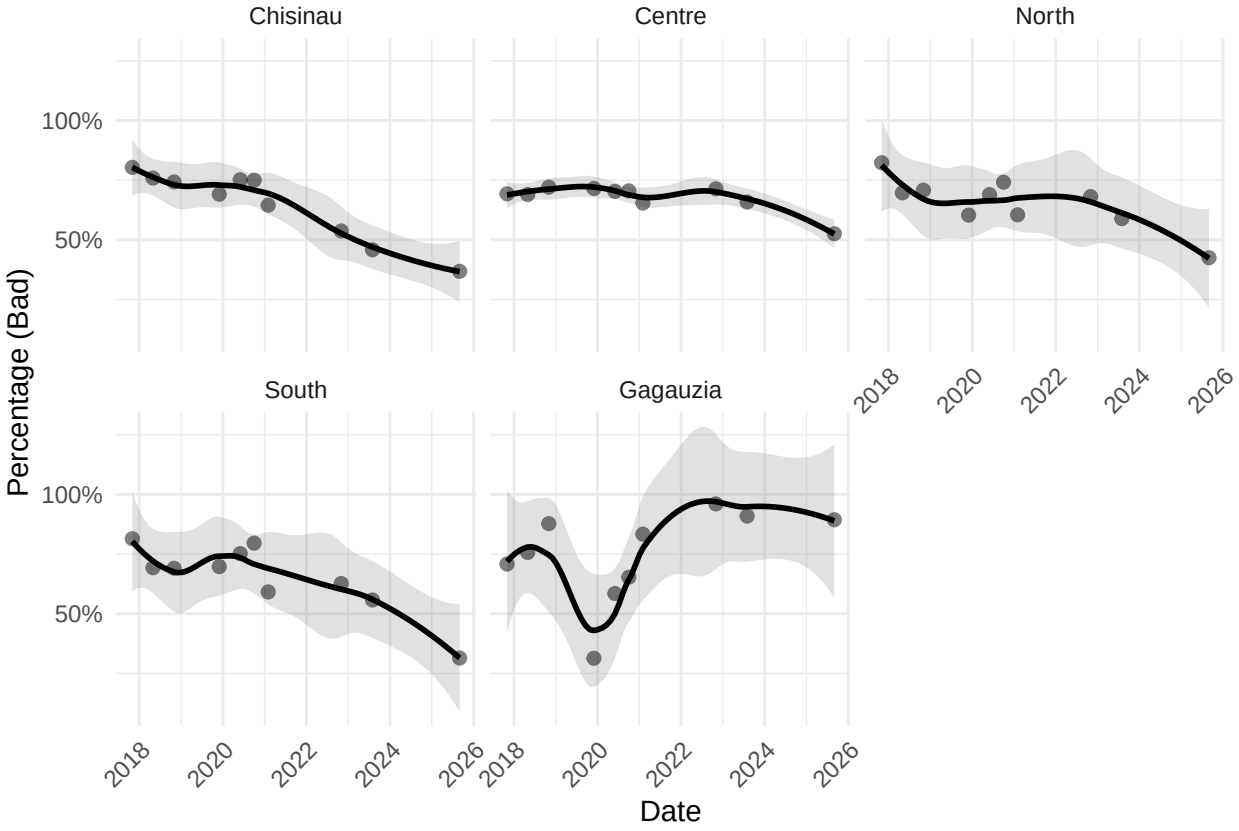
Macro-Region

```r
direction_region_df <- bop_panel %>%
  group_by(wave, date, Macro_Region) %>%
  summarise(
    pct_bad = mean(direction == "Bad", na.rm = TRUE),
    pct_good = mean(direction == "Good", na.rm = TRUE),
  )

direction_region_df <- remove_missing(direction_region_df)
```

```r
ggplot(direction_region_df, aes(x = date, y = pct_bad)) +
  geom_smooth(se = TRUE, color = "black", alpha = 0.3) +
  geom_point(size = 2, color = "black", alpha = 0.5) +
```

```
  theme_minimal() +
  scale_y_continuous(labels = scales::percent) +
  facet_wrap(~Macro_Region) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage (Bad)")
```



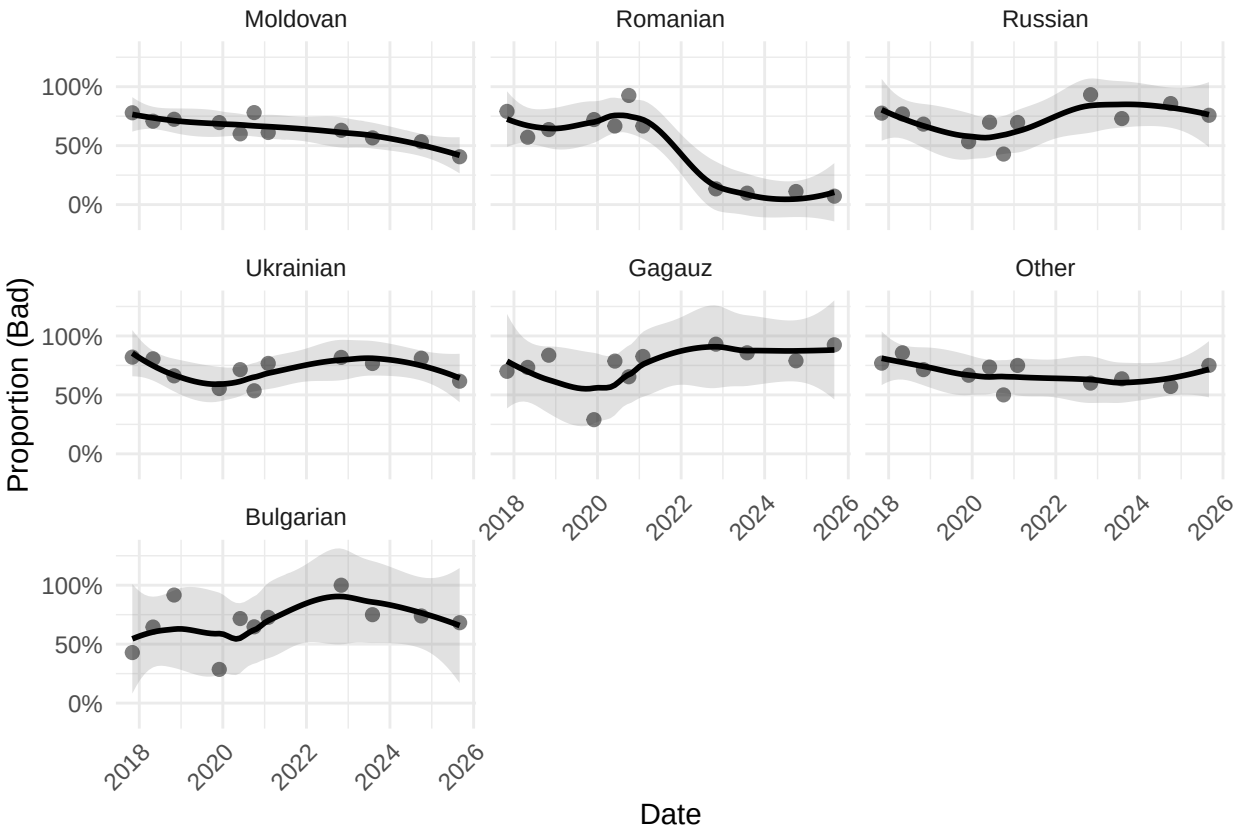
Ethnicity

```
direction_ethnicity_df <- bop_panel %>%
  group_by(wave, date, Ethnicity) %>%
  summarise(
    pct_bad = mean(direction == "Bad", na.rm = TRUE),
    pct_good = mean(direction == "Good", na.rm = TRUE),
  )

direction_ethnicity_df <- remove_missing(direction_ethnicity_df)

 direction_ethnicity_df <- direction_ethnicity_df %>%
   filter(Ethnicity != "DK/NR")
```

```r
ggplot(direction_ethnicity_df, aes(x = date, y = pct_bad)) +
  geom_smooth(se = TRUE, color = "black", alpha = 0.3) +
  geom_point(size = 2, color = "black", alpha = 0.5) +
  theme_minimal() +
  scale_y_continuous(labels = scales::percent) +
  facet_wrap(~Ethnicity) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Proportion (Bad)")
```



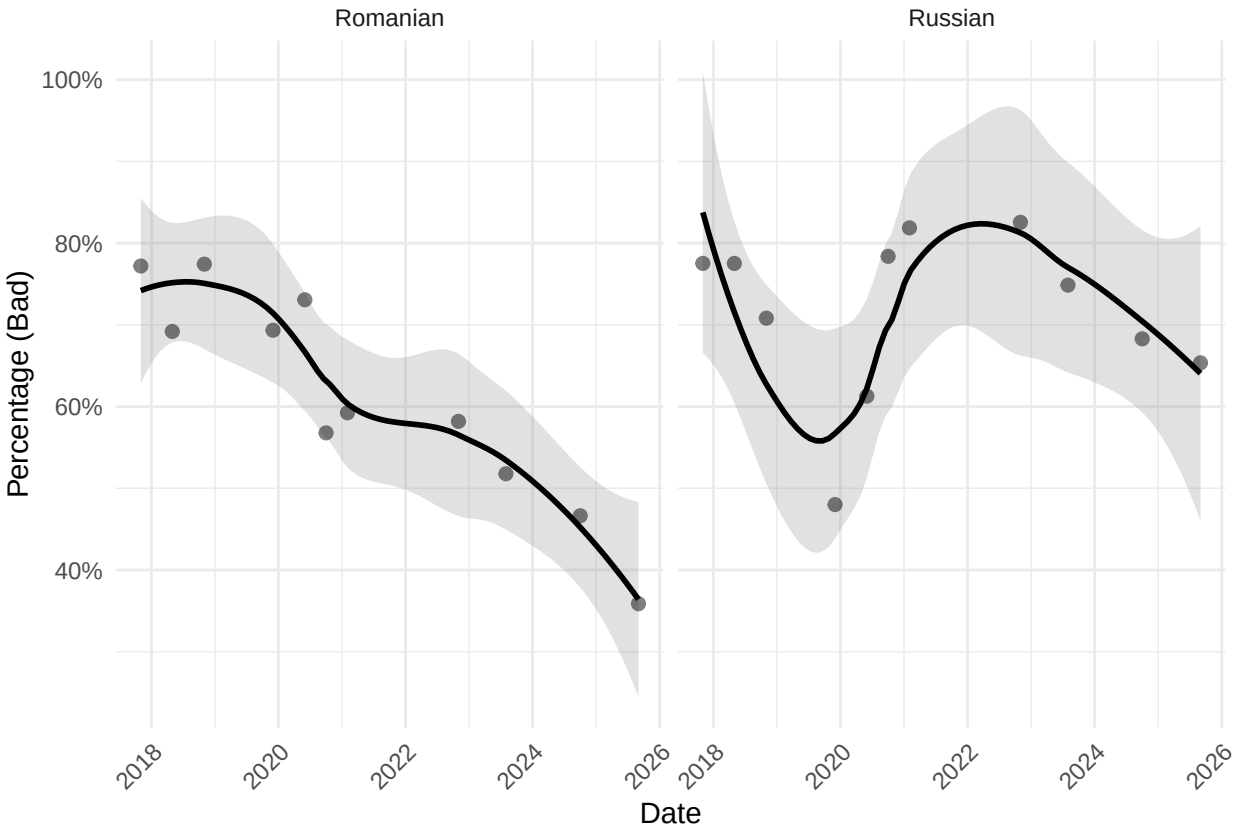
Language

```r
direction_language_df <- bop_panel %>%
  group_by(wave, date, Language_Questionnaire) %>%
  summarise(
    pct_bad = mean(direction == "Bad", na.rm = TRUE),
    pct_good = mean(direction == "Good", na.rm = TRUE),
  )

direction_language_df <- remove_missing(direction_language_df)

 direction_language_df <- direction_language_df %>%
   filter(Language_Questionnaire != "DK/NR")
```

```
ggplot(direction_language_df, aes(x = date, y = pct_bad)) +
  geom_smooth(se = TRUE, color = "black", alpha = 0.3) +
  geom_point(size = 2, color = "black", alpha = 0.5) +
  theme_minimal() +
  scale_y_continuous(labels = scales::percent) +
  facet_wrap(~Language_Questionnaire) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage (Bad)")
```



Models

```
good_models <- list(
  "Age" = glm(good ~ Age, data = bop_panel, family = binomial),
  "Gender" = glm(good ~ Gender, data = bop_panel, family = binomial),
  "Ethnicity" = glm(good ~ Ethnicity, data = bop_panel, family = binomial),
  "Language" = glm(good ~ Language_Questionnaire, data = bop_panel,
                   family = binomial),
  "Ecology" = glm(good ~ Ecology, data = bop_panel, family = binomial),
  "Education" = glm(good ~ Education, data = bop_panel, family = binomial),
  "Socioeconomic" = glm(good ~ Socioeconomic, data = bop_panel,
                        family = binomial),
  "Macro Region" = glm(good ~ Macro_Region, data = bop_panel, family = binomial),
```

```
  "War" = glm(good ~ war, data = bop_panel, family = binomial),
  "Corruption" = glm(good ~ corruption, data = bop_panel, family = binomial),
  "Interethnic" = glm(good ~ interethnic, data = bop_panel, family = binomial),
  "Poverty" = glm(good ~ poverty, data = bop_panel, family = binomial),
  "Year" = glm(good ~ year, data = bop_panel, family = binomial),
  "All" = glm(good ~ Age + Gender + Ethnicity + Language_Questionnaire +
                Ecology + Education + Socioeconomic + Macro_Region +
                war + corruption + poverty + interethnic + year,
              data = bop_panel, family = binomial)
)

modelsummary(good_models, stars = TRUE, exponentiate = TRUE)
```

```
bad_models <- list(
  "Age" = glm(bad ~ Age, data = bop_panel, family = binomial),
  "Gender" = glm(bad ~ Gender, data = bop_panel, family = binomial),
  "Ethnicity" = glm(bad ~ Ethnicity, data = bop_panel, family = binomial),
  "Language" = glm(bad ~ Language_Questionnaire, data = bop_panel,
                   family = binomial),
  "Ecology" = glm(bad ~ Ecology, data = bop_panel, family = binomial),
  "Education" = glm(bad ~ Education, data = bop_panel, family = binomial),
  "Socioeconomic" = glm(bad ~ Socioeconomic, data = bop_panel,
                        family = binomial),
  "Macro Region" = glm(bad ~ Macro_Region, data = bop_panel, family = binomial),
  "War" = glm(bad ~ war, data = bop_panel, family = binomial),
  "Corruption" = glm(bad ~ corruption, data = bop_panel, family = binomial),
  "Interethnic" = glm(bad ~ interethnic, data = bop_panel, family = binomial),
  "Poverty" = glm(bad ~ poverty, data = bop_panel, family = binomial),
  "Year" = glm(bad ~ year, data = bop_panel, family = binomial),
  "All" = glm(bad ~ Age + Gender + Ethnicity + Language_Questionnaire +
                Ecology + Education + Socioeconomic + Macro_Region +
                war + corruption + poverty + interethnic + year,
              data = bop_panel, family = binomial)
)

modelsummary(bad_models, stars = TRUE, exponentiate = TRUE)
```

Problems

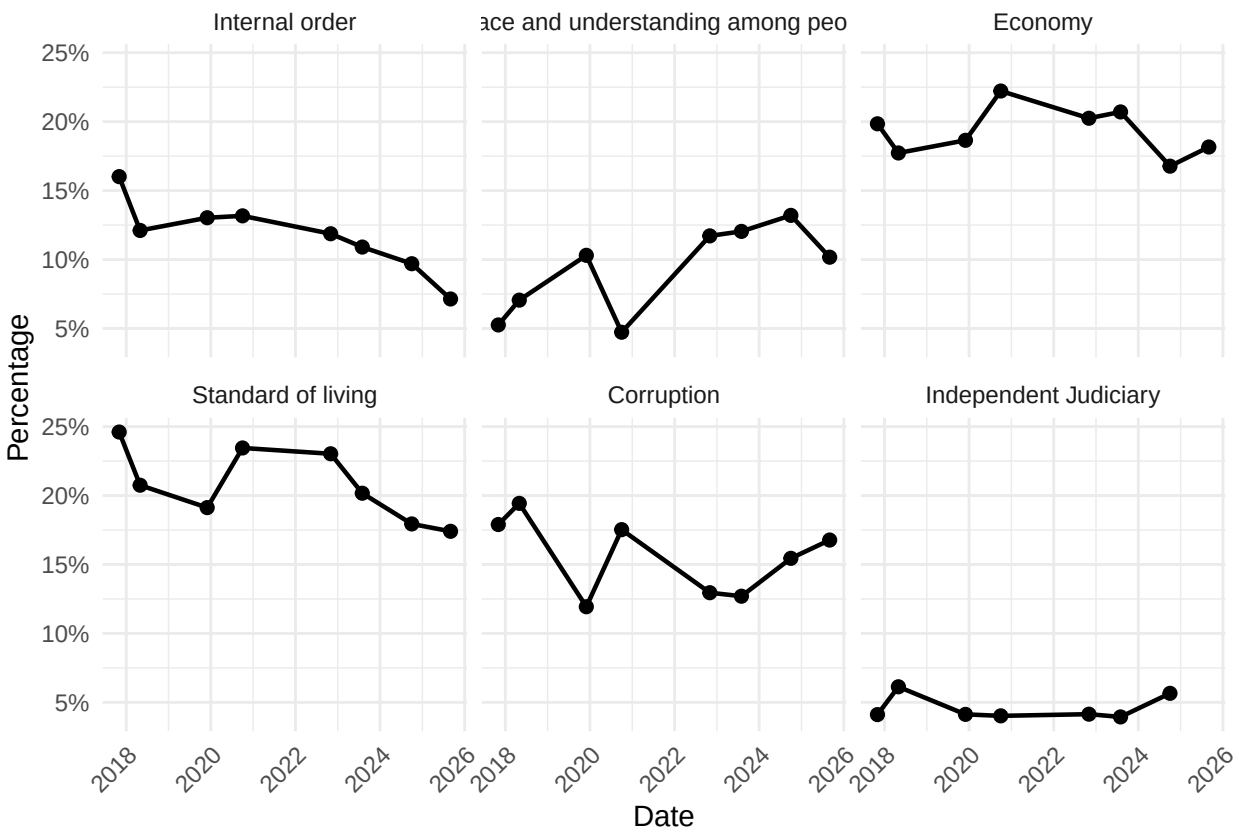
```
# mentioning problems across the 1st/2nd/3rd pref
problems_long <- bop_panel %>%
  select(wave, date, id, First_Problem, Second_Problem, Third_Problem) %>%
  pivot_longer(
    cols=c(First_Problem, Second_Problem, Third_Problem),
    names_to = "rank",
    values_to = "problem"
  )
```

	Age	Gender	Ethnicity	Language	Ecology	Education	Socioeco
(Intercept)	0.347***	0.361***	0.354***	0.427***	0.311***	0.286***	0.296
	(0.022)	(0.011)	(0.009)	(0.011)	(0.010)	(0.009)	(0.0
Age	0.999						
	(0.001)						
GenderWoman		0.876**					
		(0.036)					
EthnicityRomanian			2.731***				
			(0.229)				
EthnicityRussian			0.576***				
			(0.060)				
EthnicityUkrainian			0.669***				
			(0.055)				
EthnicityGagauz			0.578***				
			(0.067)				
EthnicityOther			0.490***				
			(0.068)				
EthnicityBulgarian			0.628***				
			(0.087)				
EthnicityDK/NR			0.419				
			(0.225)				
Language_QuestionnaireRussian				0.493***			
				(0.023)			
Ecologyrural					0.961		
					(0.043)		
Educationlow						0.848*	
						(0.057)	
Educationhigh						1.655***	
						(0.074)	
EducationDK/NR						0.292	
						(0.304)	
Socioeconomiclow							0.9
							(0.0
Socioeconomichigh							1.244
							(0.0
Macro_RegionCentre							
Macro_RegionNorth							
Macro_RegionSouth							
Macro_RegionGagauzia							
war							

	Age	Gender	Ethnicity	Language	Ecology	Education	Socioeco
(Intercept)	1.972***	1.878***	1.815***	1.510***	1.902***	2.166***	2.040
	(0.113)	(0.053)	(0.041)	(0.035)	(0.055)	(0.059)	(0.0
Age	0.999						
	(0.001)						
GenderWoman		1.004					
		(0.038)					
EthnicityRomanian			0.436***				
			(0.036)				
EthnicityRussian			1.307**				
			(0.112)				
EthnicityUkrainian			1.392***				
			(0.101)				
EthnicityGagauz			1.546***				
			(0.154)				
EthnicityOther			1.495***				
			(0.166)				
EthnicityBulgarian			1.383**				
			(0.165)				
EthnicityDK/NR			1.157				
			(0.445)				
Language_QuestionnaireRussian				1.845***			
				(0.075)			
Ecologyrural					1.061		
					(0.042)		
Educationlow						1.041	
						(0.060)	
Educationhigh						0.663***	
						(0.028)	
EducationDK/NR						1.039	
						(0.625)	
Socioeconomiclow							1.0
							(0.0
Socioeconomichigh							0.818
							(0.0
Macro_RegionCentre							
Macro_RegionNorth							
Macro_RegionSouth							
Macro_RegionGagauzia							
war							

```r
top_problems <- problems_long %>%
  filter(!is.na(problem)) %>%
  group_by(wave, date, problem) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(prop = n/sum(n))
```

```r
top_problems %>%
  filter(problem == "Economy" | problem == "Corruption" |
           problem == "Standard of living" | problem == "Internal order" |
           problem == "Corruption" | problem == "Independent Judiciary" |
           problem == "Peace and understanding among people") %>%
  ggplot(aes(x = date, y = prop)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2) +
  facet_wrap(~problem) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage")
```

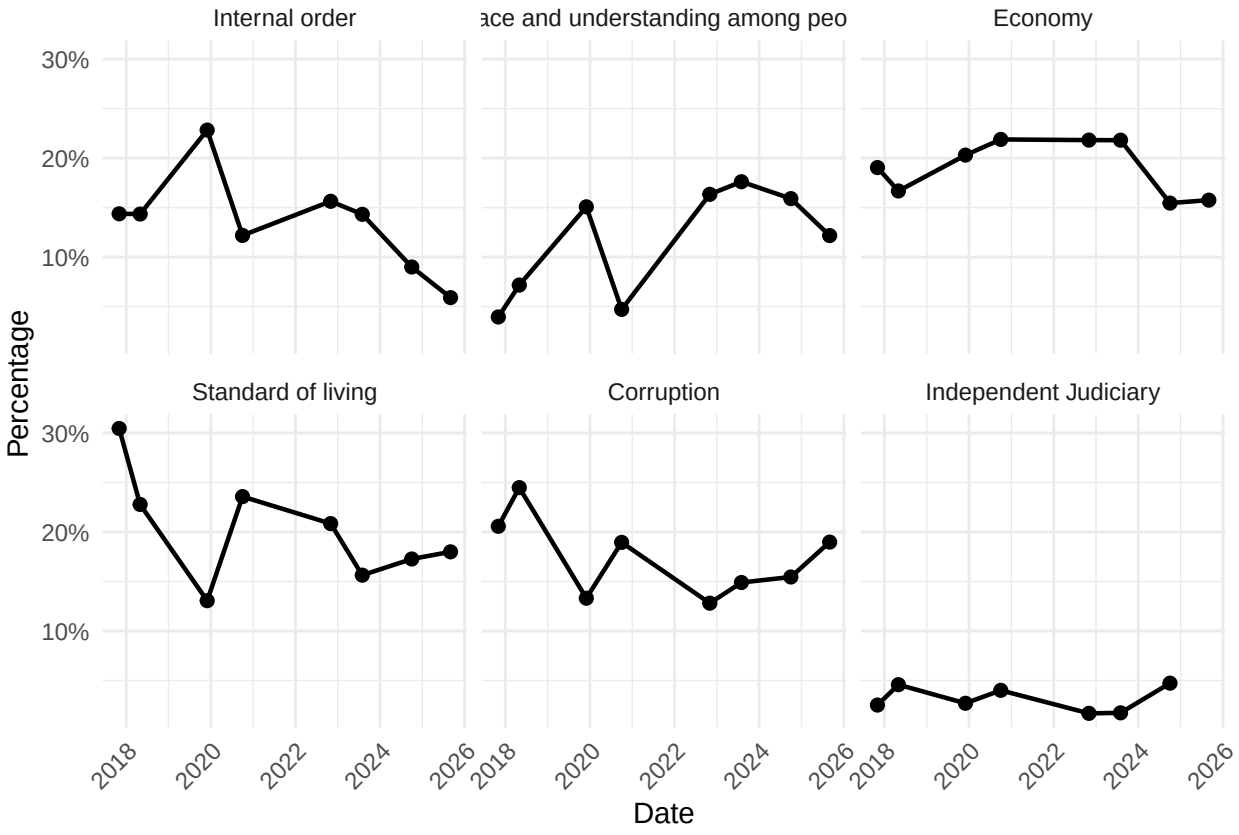



```r
# most cited first problem
first_problem <- bop_panel %>%
  group_by(wave, date, First_Problem) %>%
```

```
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(percentage = n/sum(n))

first_problem <- remove_missing(first_problem)
```

```
first_problem %>%
  filter(First_Problem == "Economy" | First_Problem == "Corruption" |
           First_Problem == "Standard of living" |
           First_Problem == "Internal order" | First_Problem == "Corruption" |
           First_Problem == "Independent Judiciary" |
           First_Problem == "Peace and understanding among people") %>%
  ggplot(aes(x = date, y = percentage)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2) +
  facet_wrap(~First_Problem) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage")
```



Worries

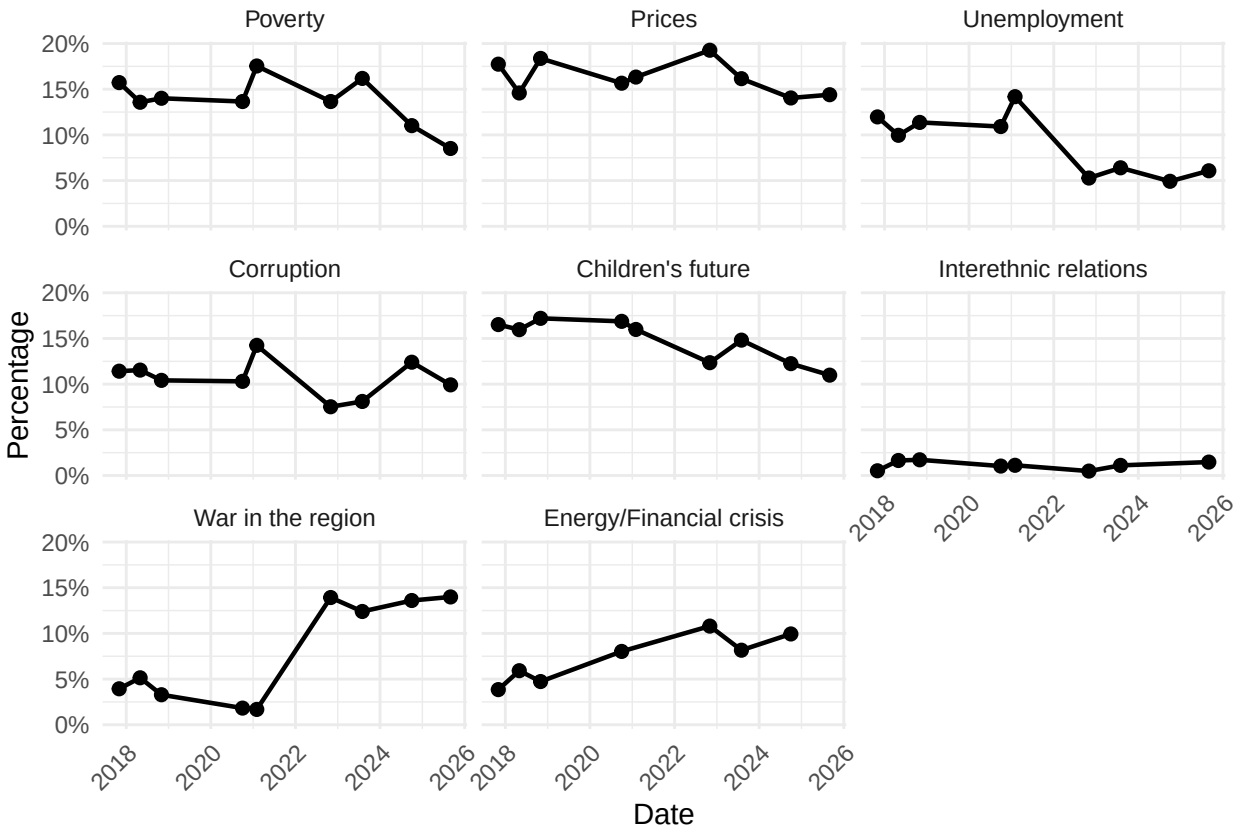
```
# merging categories that are not consistent across waves
bop_panel$First_Worry <- fct_collapse(bop_panel$First_Worry ,
                              "Other"    = c("Other", "Housing", "Education",
                                             "Lack of social services",
                                             "Climate change", "Emigration",
                                             "Low salary/pension"),
                              "Energy/Financial crisis" = c("Energy crisis",
                                                            "Financial crisis"))
bop_panel$Second_Worry <- fct_collapse(bop_panel$Second_Worry ,
                              "Other"    = c("Other", "Housing", "Education",
                                             "Lack of social services",
                                             "Climate change", "Emigration",
                                             "Low salary/pension"),
                              "Energy/Financial crisis" = c("Energy crisis",
                                                            "Financial crisis"))
bop_panel$Third_Worry <- fct_collapse(bop_panel$Third_Worry ,
                              "Other"    = c("Other", "Housing", "Education",
                                             "Lack of social services",
                                             "Climate change", "Emigration",
                                             "Low salary/pension"),
                              "Energy/Financial crisis" = c("Energy crisis",
                                                            "Financial crisis"))
```

```
# mentioning problems across the 1st/2nd/3rd pref
worries_long <- bop_panel %>%
  select(wave, date, id, First_Worry, Second_Worry, Third_Worry) %>%
  pivot_longer(
    cols=c(First_Worry, Second_Worry, Third_Worry),
    names_to = "rank",
    values_to = "worries"
  )
```

```
top_worries <- worries_long %>%
  filter(!is.na(worries)) %>%
  group_by(wave, date, worries) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(prop = n/sum(n))
```

```
top_worries %>%
  filter(worries == "Poverty" | worries == "Prices" | worries == "Unemployment" |
           worries == "Corruption" | worries == "Children's future" |
           worries == "Interethnic relations" | worries == "War in the region" |
           worries == "Energy/Financial crisis") %>%
  ggplot(aes(x = date, y = prop)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2) +
  facet_wrap(~worries) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
```

```
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage")
```

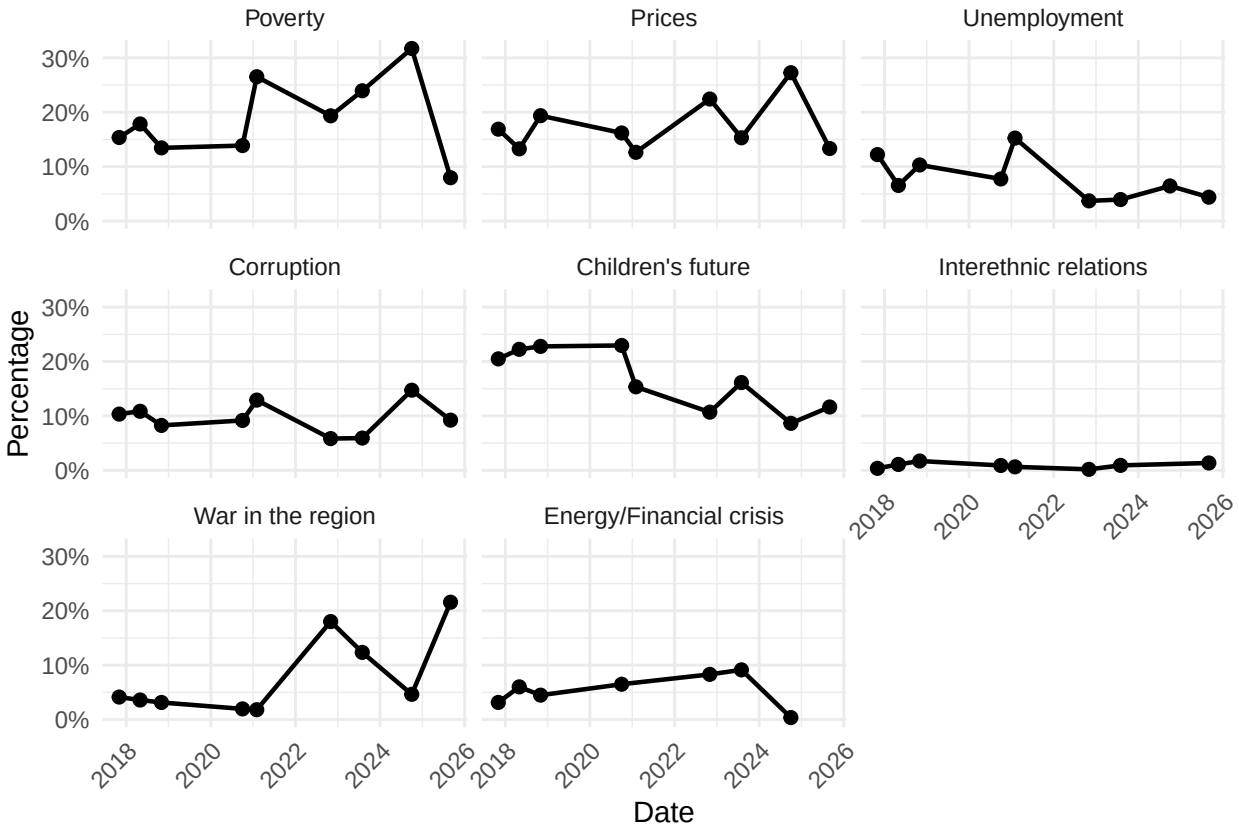



```
# most cited first problem
first_worry <- bop_panel %>%
  group_by(wave, date, First_Worry) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(percentage = n/sum(n))

first_worry <- remove_missing(first_worry)
```

```
first_worry %>%
  filter(First_Worry == "Poverty" | First_Worry == "Prices" |
           First_Worry == "Unemployment" | First_Worry == "Corruption" |
           First_Worry == "Children's future" |
           First_Worry == "Interethnic relations" |
           First_Worry == "War in the region" |
           First_Worry == "Energy/Financial crisis") %>%
  ggplot(aes(x = date, y = percentage)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2) +
  facet_wrap(~First_Worry) +
  scale_y_continuous(labels = scales::percent) +
```

```
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage")
```



Media

```
bop_panel$Trust_Media <- fct_collapse(bop_panel$Trust_Media ,
                              "DK/NR"    = c("DK/NR", "None"),
                              "Internet" = c("Internet", "Social media"))
bop_panel$Important_Media <- fct_collapse(bop_panel$Important_Media ,
                              "DK/NR"    = c("DK/NR", "None"),
                              "Internet" = c("Internet", "Social media"))
bop_panel$Second_Media <- fct_collapse(bop_panel$Second_Media ,
                              "DK/NR"    = c("DK/NR", "None"),
                              "Internet" = c("Internet", "Social media"))
```

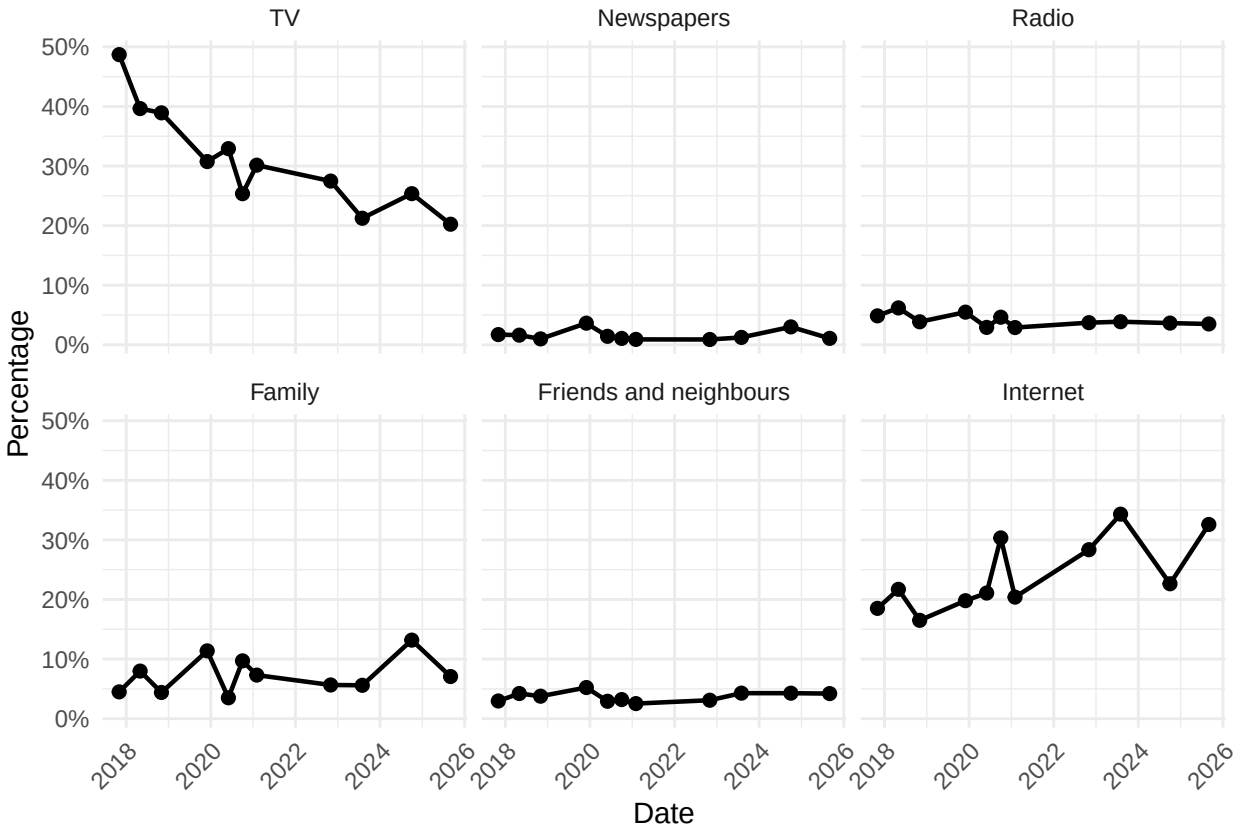
```
bop_panel$top_TV <- ifelse(bop_panel$Important_Media == "TV", 1, 0)
bop_panel$top_internet <- ifelse(bop_panel$Important_Media == "Internet", 1, 0)
bop_panel$trust_TV <- ifelse(bop_panel$Trust_Media == "TV", 1, 0)
bop_panel$trust_internet <- ifelse(bop_panel$Trust_Media == "Internet", 1, 0)
```

Most trusted

```
trusted_media <- bop_panel %>%
  group_by(wave, date, Trust_Media) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(percentage = n/sum(n))

trusted_media <- remove_missing(trusted_media)
```

```
trusted_media %>%
  filter(Trust_Media == "Internet" | Trust_Media == "TV" |
           Trust_Media == "Radio" | Trust_Media == "Newspapers" |
           Trust_Media == "Family" | Trust_Media == "Friends and neighbours") %>%
  ggplot(aes(x = date, y = percentage)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2) +
  facet_wrap(~Trust_Media) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage")
```

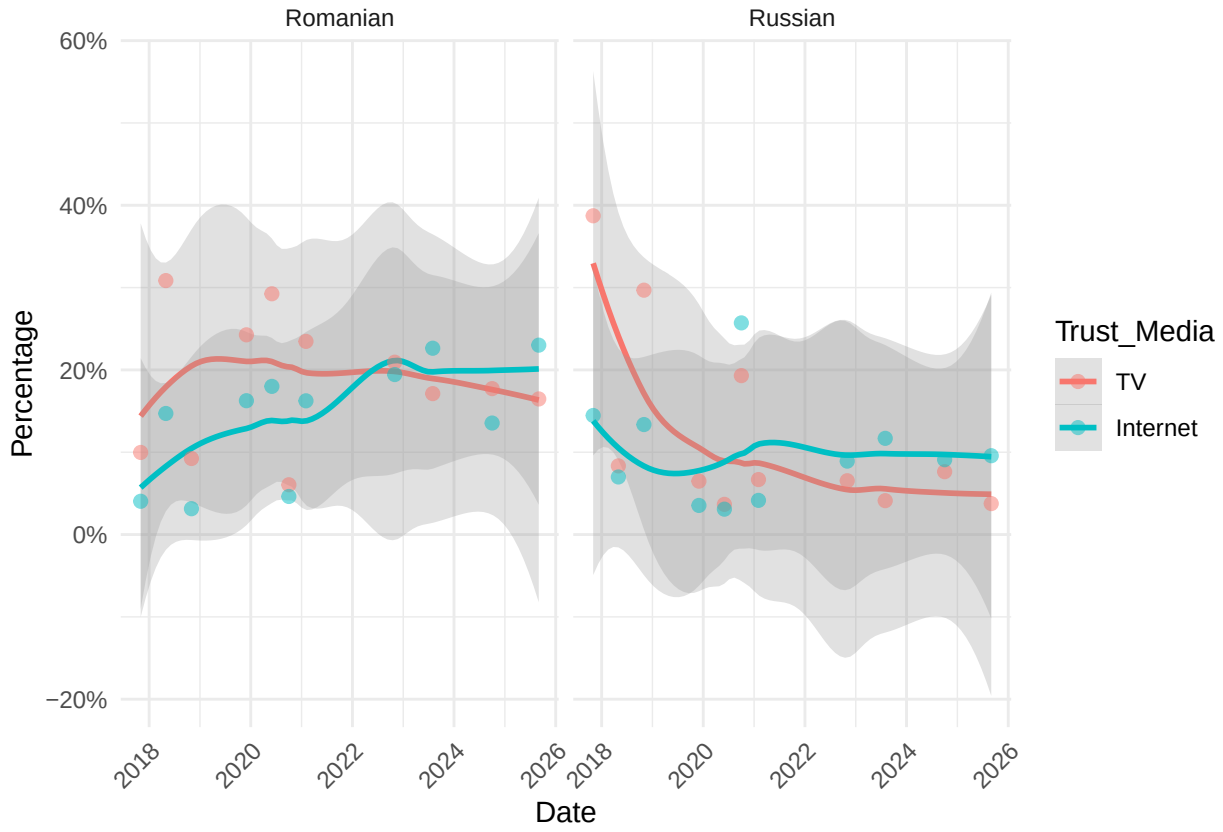


```r
# most trusted form of media by language of survey

trusted_media_lang <- bop_panel %>%
  group_by(wave, date, Trust_Media, Language_Questionnaire) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(percentage = n/sum(n))

trusted_media_lang <- remove_missing(trusted_media_lang)
```

```r
trusted_media_lang %>%
  filter(Trust_Media == "Internet" | Trust_Media == "TV") %>%
  ggplot(aes(x = date, y = percentage, color = Trust_Media)) +
  geom_smooth(se = TRUE, alpha = 0.3) +
  geom_point(size = 2, alpha = 0.5) +
  facet_wrap(~Language_Questionnaire) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage")
```



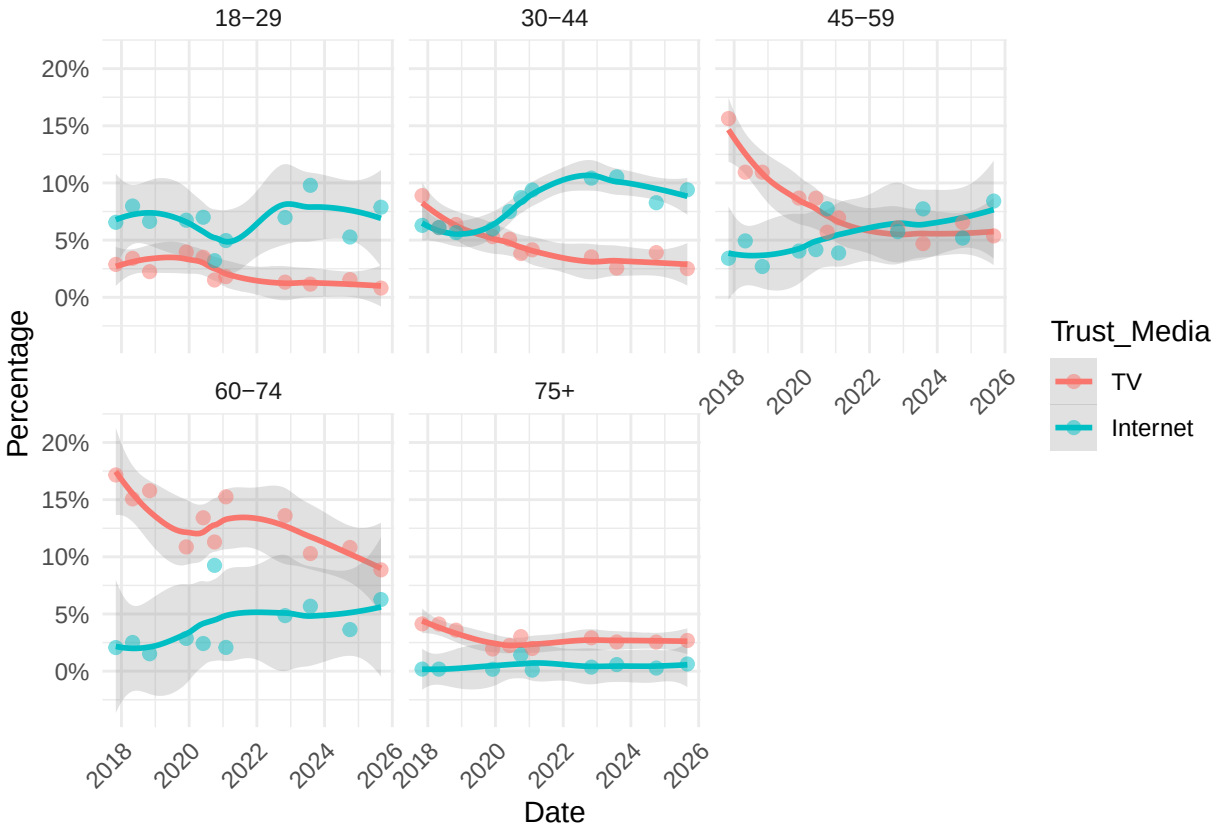
Language

```r
# most trusted form of media by language of survey

trusted_media_age <- bop_panel %>%
  group_by(wave, date, Trust_Media, Age.f) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(percentage = n/sum(n))

trusted_media_age <- remove_missing(trusted_media_age)
```

```r
trusted_media_age %>%
  filter(Trust_Media == "Internet" | Trust_Media == "TV") %>%
  ggplot(aes(x = date, y = percentage, color = Trust_Media)) +
  geom_smooth(se = TRUE, alpha = 0.3) +
  geom_point(size = 2, alpha = 0.5) +
  facet_wrap(~Age.f) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage")
```



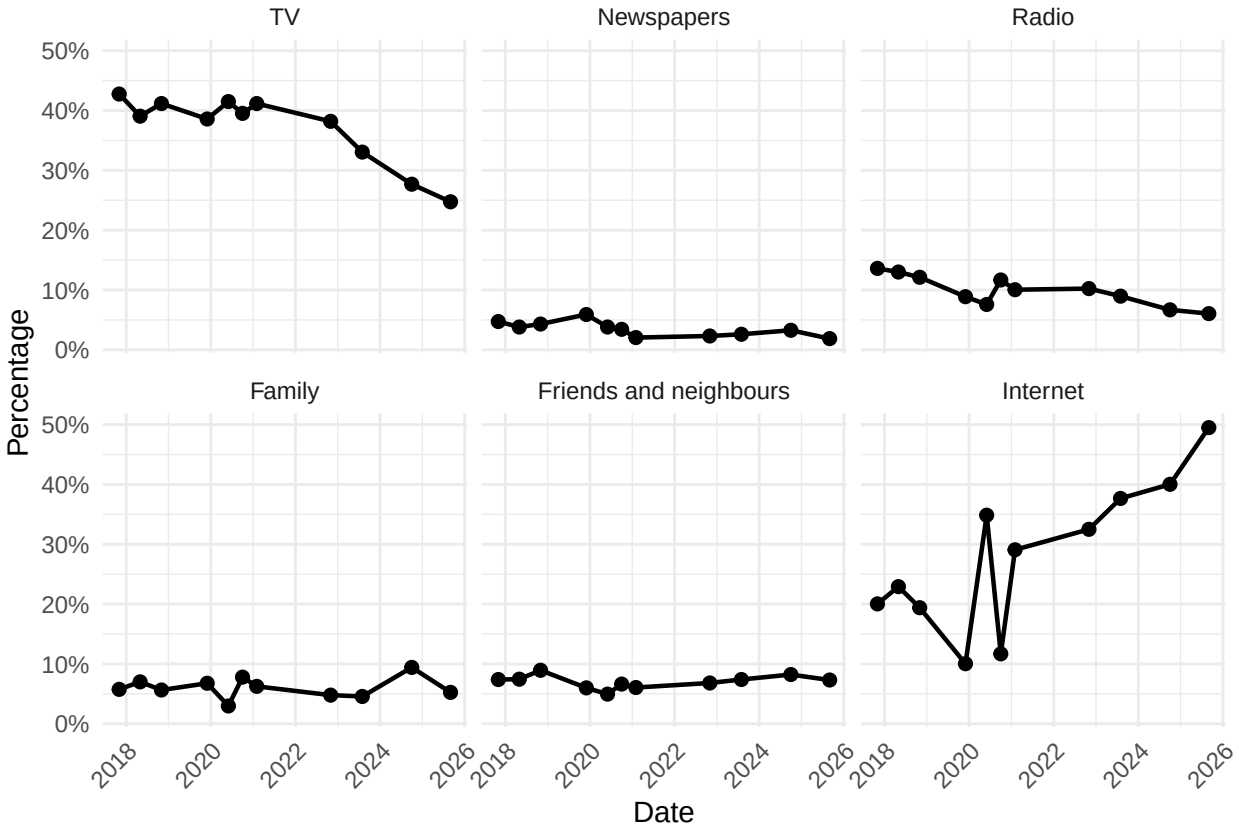
Age

Most important media

```
# important media mentioned across the 1st/2nd pref
media_pref_long <- bop_panel %>%
  select(wave, date, id, Important_Media, Second_Media) %>%
  pivot_longer(
    cols=c(Important_Media, Second_Media),
    names_to = "rank",
    values_to = "media"
  )

top_media_pref <- media_pref_long %>%
  filter(!is.na(media)) %>%
  group_by(wave, date, media) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(prop = n/sum(n))
```

```
top_media_pref %>%
  filter(media == "Internet" | media == "TV" | media == "Radio" | media == "Newspapers" | media == "Fam
  ggplot(aes(x = date, y = prop)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2) +
  facet_wrap(~media) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage")
```



Language

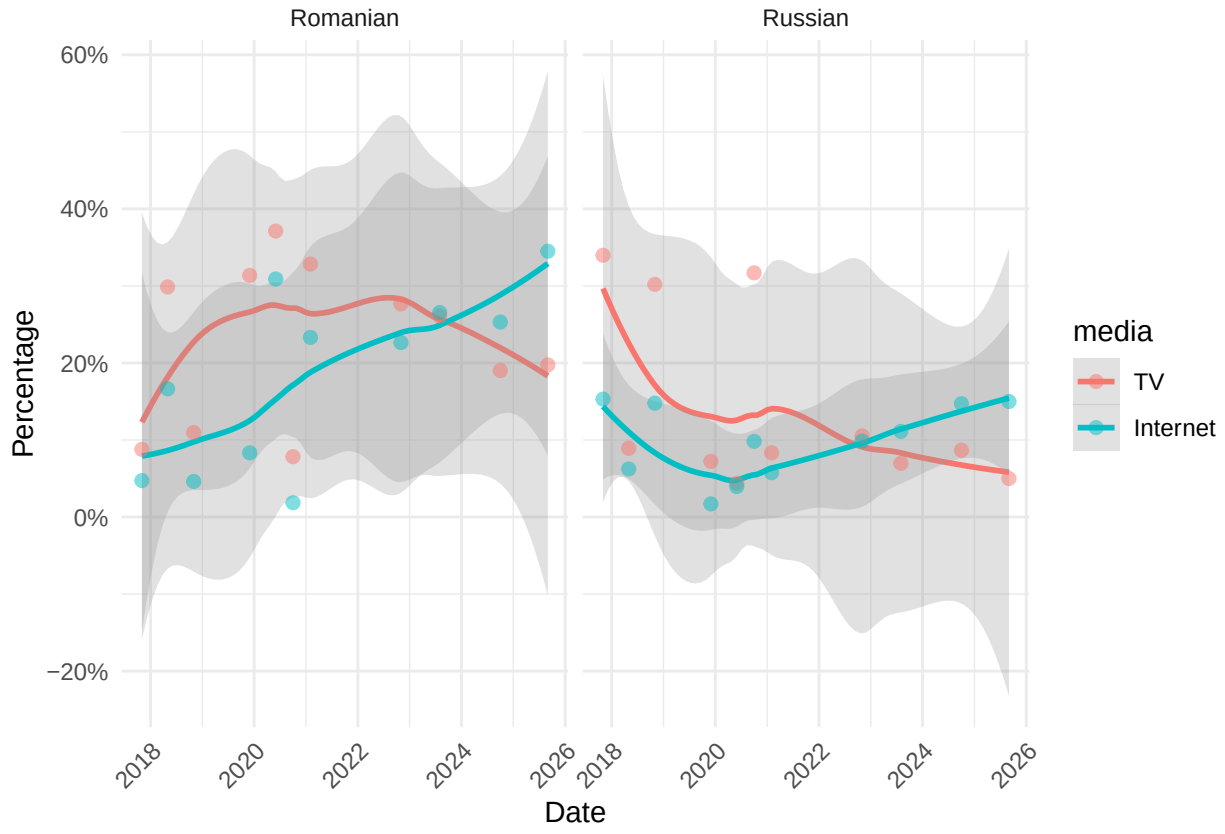
```
media_pref_long_lang <- bop_panel %>%
  select(wave, date, id, Language_Questionnaire, Important_Media, Second_Media) %>%
  pivot_longer(
    cols=c(Important_Media, Second_Media),
    names_to = "rank",
    values_to = "media"
  )

top_media_pref_lang <- media_pref_long_lang %>%
  filter(!is.na(media)) %>%
  group_by(wave, date, Language_Questionnaire, media) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(prop = n/sum(n))

top_media_pref_lang <- remove_missing(top_media_pref_lang)
```

```
top_media_pref_lang %>%
  filter(media == "Internet" | media == "TV") %>%
  ggplot(aes(x = date, y = prop, color = media)) +
  geom_smooth(se = TRUE, alpha = 0.3) +
  geom_point(size = 2, alpha = 0.5) +
  facet_wrap(~Language_Questionnaire) +
  scale_y_continuous(labels = scales::percent) +
```

```
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage")
```

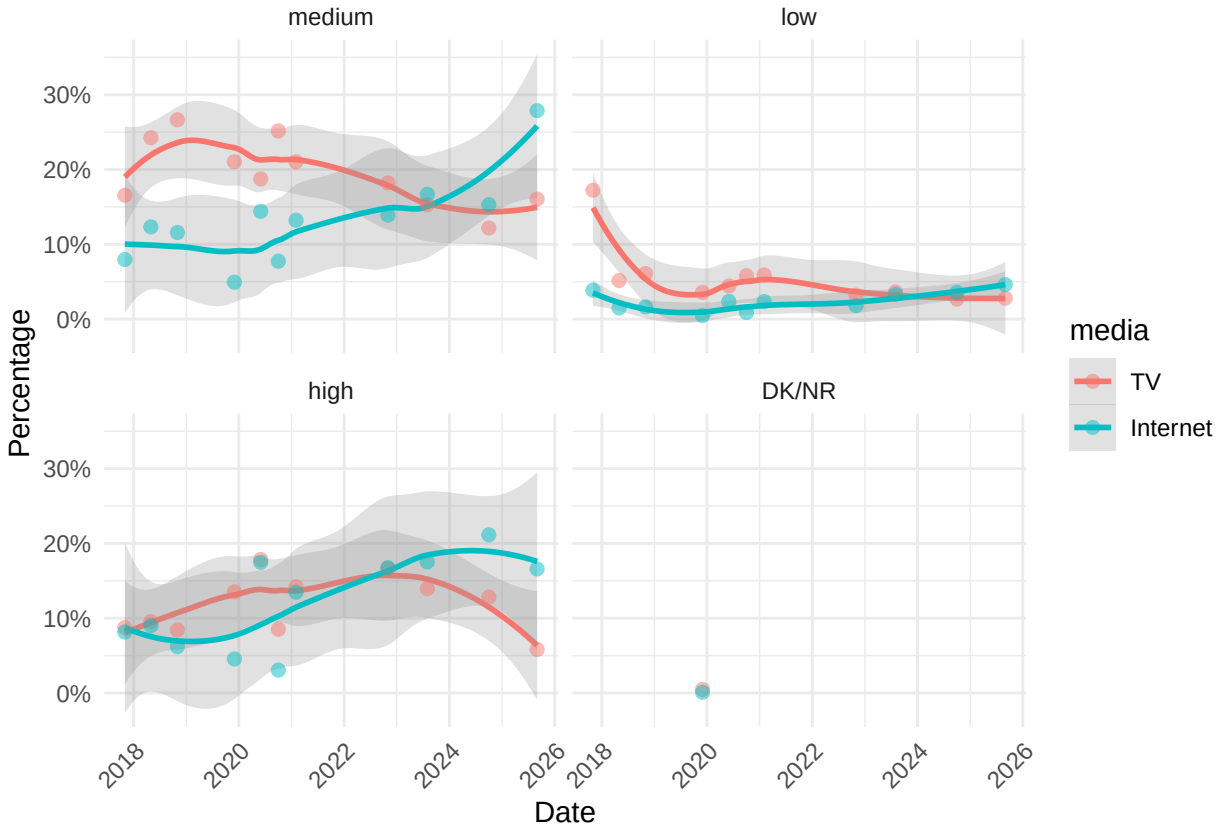



```
media_pref_long_edu <- bop_panel %>%
  select(wave, date, id, Education, Important_Media, Second_Media) %>%
  pivot_longer(
    cols=c(Important_Media, Second_Media),
    names_to = "rank",
    values_to = "media"
  )

top_media_pref_edu <- media_pref_long_edu %>%
  filter(!is.na(media)) %>%
  group_by(wave, date, Education, media) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(prop = n/sum(n))

top_media_pref_edu <- remove_missing(top_media_pref_edu)
```

```r
top_media_pref_edu %>%
  filter(media == "Internet" | media == "TV") %>%
  ggplot(aes(x = date, y = prop, color = media)) +
  geom_smooth(se = TRUE, alpha = 0.3) +
  geom_point(size = 2, alpha = 0.5) +
  facet_wrap(~Education) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Percentage")
```



Education

Models

```r
top_TV_models <- list(
  "Age" = glm(top_TV ~ Age, data = bop_panel, family = binomial),
  "Gender" = glm(top_TV ~ Gender, data = bop_panel, family = binomial),
  "Ethnicity" = glm(top_TV ~ Ethnicity, data = bop_panel, family = binomial),
  "Language" = glm(top_TV ~ Language_Questionnaire, data = bop_panel,
                   family = binomial),
  "Ecology" = glm(top_TV ~ Ecology, data = bop_panel, family = binomial),
```

```
  "Education" = glm(top_TV ~ Education, data = bop_panel, family = binomial),
  "Socioeconomic" = glm(top_TV ~ Socioeconomic, data = bop_panel,
                        family = binomial),
  "Macro Region" = glm(top_TV ~ Macro_Region, data = bop_panel, family = binomial),
  "Year" = glm(top_TV ~ year, data = bop_panel, family = binomial),
  "All" = glm(top_TV ~ Age + Gender + Ethnicity + Language_Questionnaire +
                Ecology + Education + Socioeconomic + Macro_Region + year,
              data = bop_panel, family = binomial)
)

modelsummary(top_TV_models, stars = TRUE, exponentiate = TRUE, coef_map = coef_model)
```

```
top_internet_models <- list(
  "Age" = glm(top_internet ~ Age, data = bop_panel, family = binomial),
  "Gender" = glm(top_internet ~ Gender, data = bop_panel, family = binomial),
  "Ethnicity" = glm(top_internet ~ Ethnicity, data = bop_panel, family = binomial),
  "Language" = glm(top_internet ~ Language_Questionnaire, data = bop_panel,
                   family = binomial),
  "Ecology" = glm(top_internet ~ Ecology, data = bop_panel, family = binomial),
  "Education" = glm(top_internet ~ Education, data = bop_panel, family = binomial),
  "Socioeconomic" = glm(top_internet ~ Socioeconomic, data = bop_panel,
                        family = binomial),
  "Macro Region" = glm(top_internet ~ Macro_Region, data = bop_panel, family = binomial),
  "Year" = glm(top_internet ~ year, data = bop_panel, family = binomial),
  "All" = glm(top_internet ~ Age + Gender + Ethnicity + Language_Questionnaire +
                Ecology + Education + Socioeconomic + Macro_Region + year,
              data = bop_panel, family = binomial)
)

modelsummary(top_internet_models, stars = TRUE, exponentiate = TRUE, coef_map = coef_model)
```

Domestic politics

```
bop_panel$year <- year(bop_panel$date)
bop_panel$year.f <- as.factor(bop_panel$year)
bop_panel$wave <- as.factor(bop_panel$wave)

bop_panel$Sandu <- as.numeric(bop_panel$Sandu)
bop_panel$Dodon <- as.numeric(bop_panel$Dodon)
bop_panel$Sor <- as.numeric(bop_panel$Sor)
bop_panel$PAS <- as.numeric(bop_panel$PAS)
bop_panel$PSRM <- as.numeric(bop_panel$PSRM)
bop_panel$Putin <- as.numeric(bop_panel$Putin)
bop_panel$Zelenskyy <- as.numeric(bop_panel$Zelenskyy)
```

```
domestic_bop <- bop_panel %>%
  group_by(wave, date) %>%
  summarise(
    Sandu_mean = mean(Sandu, na.rm = TRUE),
```

	Age	Gender	Ethnicity	Language	Ecology	Education	Socioeconomic	M
Age (per year)	1.042***							
	(0.001)							
Woman		1.180***						
		(0.043)						
Romanian			0.982					
			(0.081)					
Russian			1.040					
			(0.081)					
Ukrainian			1.155*					
			(0.076)					
Gagauz			0.864+					
			(0.076)					
Bulgarian			0.975					
			(0.105)					
Russian questionnaire				1.101**				
				(0.041)				
Rural					1.541***			
					(0.058)			
Education: low						1.162**		
						(0.062)		
Education: high						0.628***		
						(0.025)		
Socioeconomic: low							1.414***	
							(0.065)	
Socioeconomic: high							0.643***	
							(0.031)	
Centre								
North								
South								
Gagauzia								
Year								
Num.Obs.	12 520	12 520	12 520	12 514	11 426	12 475	11 113	
AIC	15 871.5	17 303.1	17 293.5	17 308.3	15 697.5	17 088.6	15 117.8	
BIC	15 886.4	17 318.0	17 352.9	17 323.2	15 712.2	17 118.3	15 139.7	
Log.Lik.	−7933.770	−8649.542	−8638.731	−8652.154	−7846.756	−8540.307	−7555.893	
F	1291.241	20.993	5.462	6.812		58.525		
RMSE	0.47	0.50	0.50	0.50	0.50	0.50	0.49	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

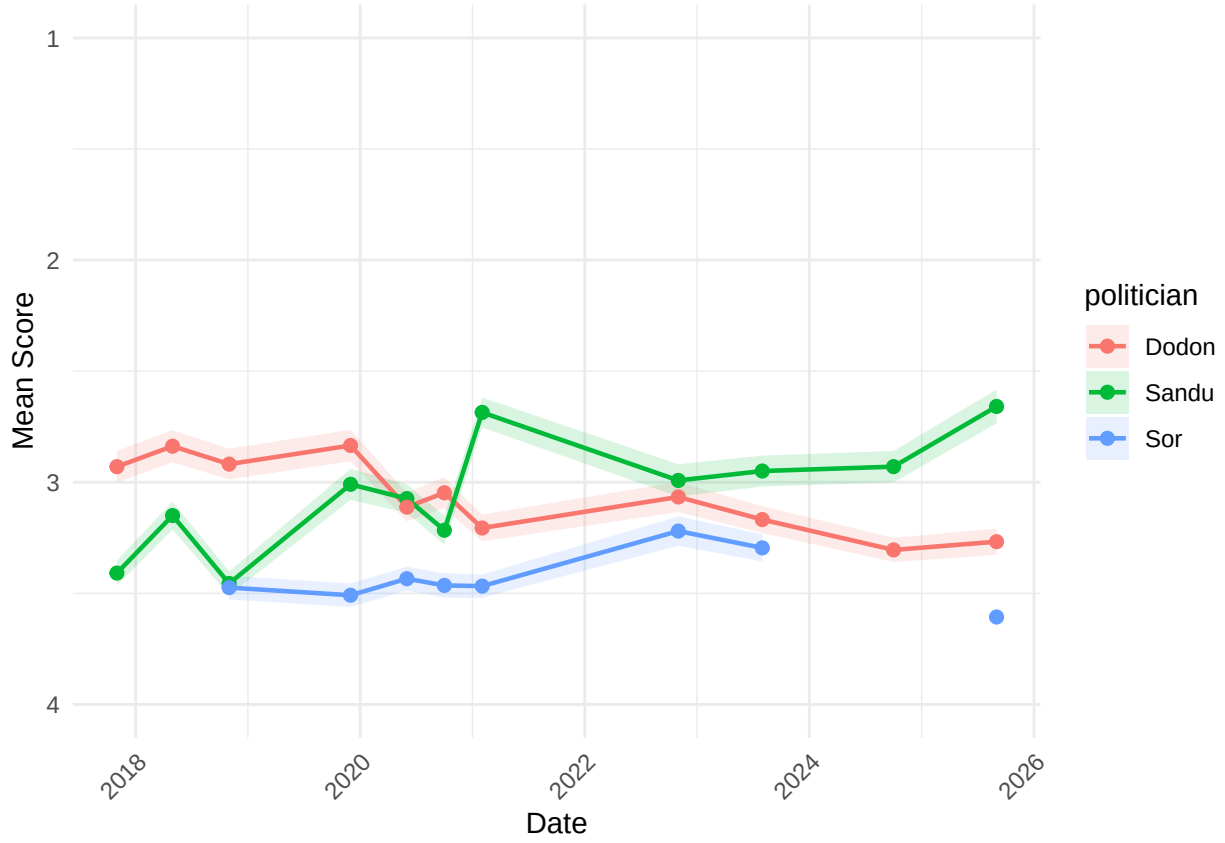
	Age	Gender	Ethnicity	Language	Ecology	Education	Socioeconomic	M
Age (per year)	0.953***							
	(0.001)							
Woman		0.873***						
		(0.034)						
Romanian			1.258**					
			(0.111)					
Russian			1.134					
			(0.096)					
Ukrainian			1.041					
			(0.075)					
Gagauz			1.121					
			(0.107)					
Bulgarian			1.396**					
			(0.157)					
Russian questionnaire				0.767***				
				(0.031)				
Rural					0.566***			
					(0.024)			
Education: low						0.549***		
						(0.038)		
Education: high						1.915***		
						(0.081)		
Socioeconomic: low							0.477***	
							(0.026)	
Socioeconomic: high							1.340***	
							(0.068)	
Centre								
North								
South								
Gagauzia								
Year								
Num.Obs.	12 520	12 520	12 520	12 514	11 426	12 475	11 113	
AIC	13 622.9	15 188.5	15 097.8	15 153.7	13 541.5	14 695.7	12 878.9	
BIC	13 637.7	15 203.4	15 157.3	15 168.6	13 556.2	14 725.4	12 900.8	
Log.Lik.	−6809.436	−7592.267	−7540.913	−7574.852	−6768.770	−7343.832	−6436.447	
F	1373.014	11.954	16.735	42.056		138.001		
RMSE	0.43	0.46	0.45	0.46	0.45	0.45	0.44	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

```
    Sandu_sd = sd(Sandu, na.rm = TRUE),
    Sandu_n = sum(!is.na(Sandu)),
    Sandu_se = Sandu_sd / sqrt(Sandu_n),
    Sandu_lower = Sandu_mean - 1.96 * Sandu_se,
    Sandu_upper = Sandu_mean + 1.96 * Sandu_se,
    Dodon_mean = mean(Dodon, na.rm = TRUE),
    Dodon_sd = sd(Dodon, na.rm = TRUE),
    Dodon_n = sum(!is.na(Dodon)),
    Dodon_se = Dodon_sd / sqrt(Dodon_n),
    Dodon_lower = Dodon_mean - 1.96 * Dodon_se,
    Dodon_upper = Dodon_mean + 1.96 * Dodon_se,
    Sor_mean = mean(Sor, na.rm = TRUE),
    Sor_sd = sd(Sor, na.rm = TRUE),
    Sor_n = sum(!is.na(Sor)),
    Sor_se = Sor_sd / sqrt(Sor_n),
    Sor_lower = Sor_mean - 1.96 * Sor_se,
    Sor_upper = Sor_mean + 1.96 * Sor_se
  )
```

```
domestic_bop_long <- domestic_bop %>%
  pivot_longer(
    cols = c(Sandu_mean, Dodon_mean, Sor_mean, Sandu_lower, Sandu_upper, Dodon_lower, Dodon_upper, Sor_
    names_to = c("politician", ".value"),
    names_pattern = "(Sandu|Dodon|Sor)_(.*)"
  )
```

```
ggplot(domestic_bop_long, aes(x = date, y = mean, color = politician,
                              group = politician)) +
  geom_ribbon(aes(ymin = lower, ymax = upper, fill = politician),
              alpha = 0.15, color = NA) +
  geom_line(linewidth = 0.8) +
  scale_y_reverse(limits = c(1,4)) +
  geom_point(size = 2) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Mean Score")
```

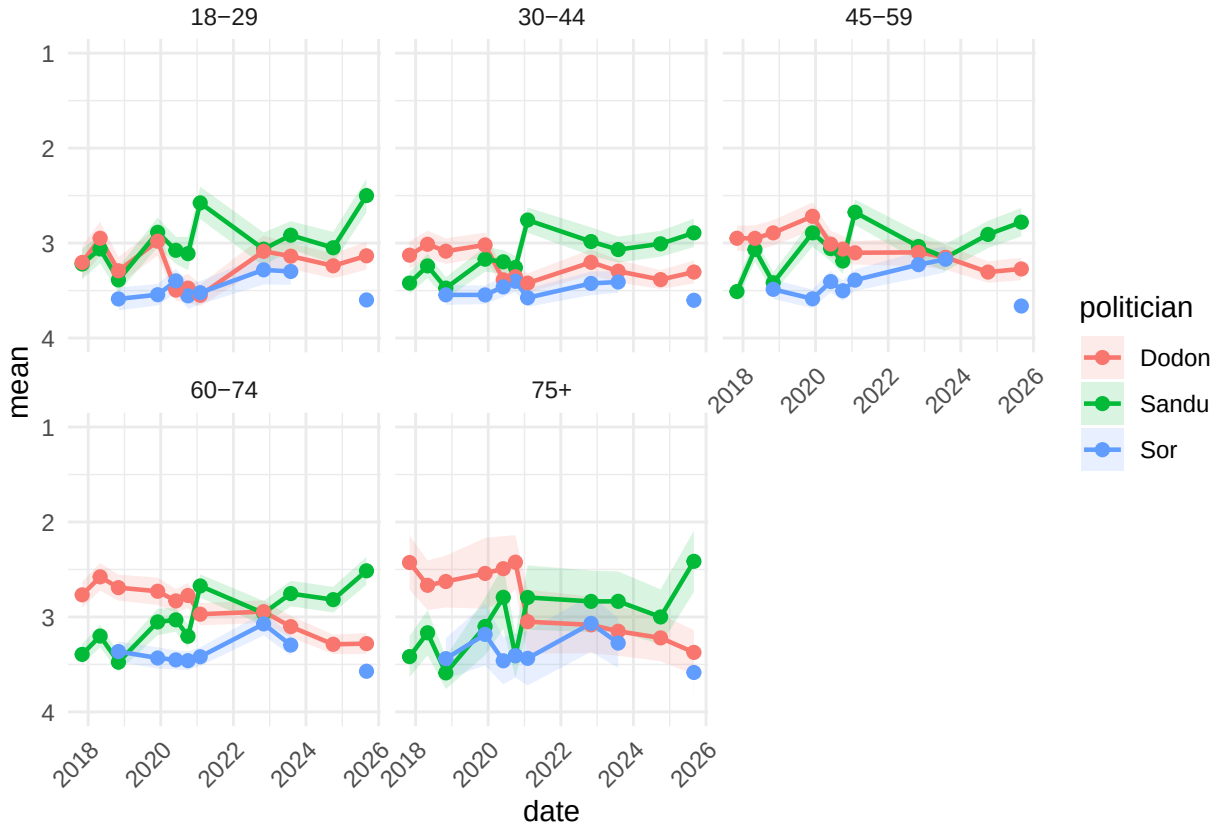


Age

```
domestic_bop_age <- bop_panel %>%
  group_by(wave, date, Age.f) %>%
  summarise(
    Sandu_mean = mean(Sandu, na.rm = TRUE),
    Sandu_sd = sd(Sandu, na.rm = TRUE),
    Sandu_n = sum(!is.na(Sandu)),
    Sandu_se = Sandu_sd / sqrt(Sandu_n),
    Sandu_lower = Sandu_mean - 1.96 * Sandu_se,
    Sandu_upper = Sandu_mean + 1.96 * Sandu_se,
    Dodon_mean = mean(Dodon, na.rm = TRUE),
    Dodon_sd = sd(Dodon, na.rm = TRUE),
    Dodon_n = sum(!is.na(Dodon)),
    Dodon_se = Dodon_sd / sqrt(Dodon_n),
    Dodon_lower = Dodon_mean - 1.96 * Dodon_se,
    Dodon_upper = Dodon_mean + 1.96 * Dodon_se,
    Sor_mean = mean(Sor, na.rm = TRUE),
    Sor_sd = sd(Sor, na.rm = TRUE),
    Sor_n = sum(!is.na(Sor)),
    Sor_se = Sor_sd / sqrt(Sor_n),
    Sor_lower = Sor_mean - 1.96 * Sor_se,
    Sor_upper = Sor_mean + 1.96 * Sor_se
  )
```

```
domestic_bop_age_long <- domestic_bop_age %>%
  pivot_longer(
    cols = c(Sandu_mean, Dodon_mean, Sor_mean, Sandu_lower, Sandu_upper, Dodon_lower, Dodon_upper, Sor_
    names_to = c("politician", ".value"),
    names_pattern = "(Sandu|Dodon|Sor)_(.*)"
  )
```

```
ggplot(domestic_bop_age_long, aes(x = date, y = mean, color = politician,
                                  group = politician)) +
  geom_ribbon(aes(ymin = lower, ymax = upper, fill = politician),
              alpha = 0.15, color = NA) +
  geom_line(linewidth = 0.8) +
  scale_y_reverse(limits = c(1,4)) +
  geom_point(size = 2) +
  facet_wrap(~Age.f) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Language

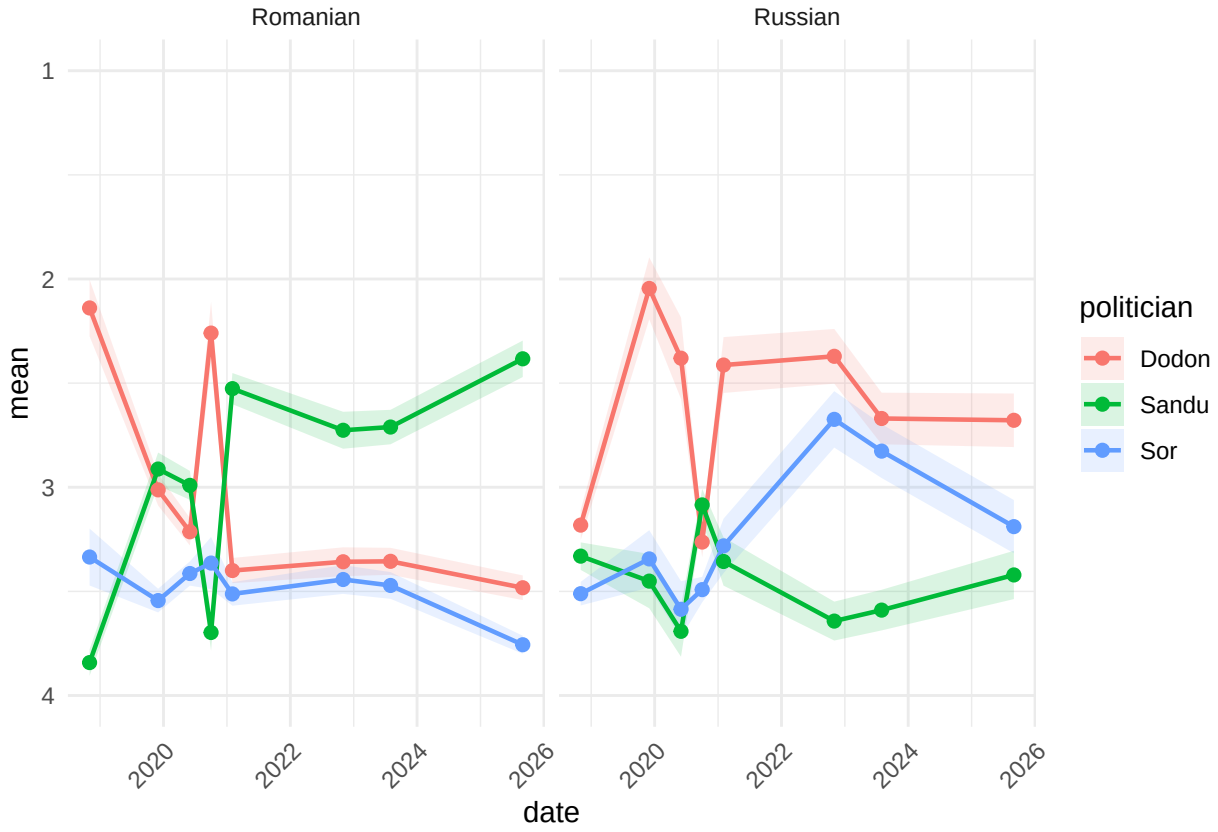
```
domestic_bop_language <- bop_panel %>%
  group_by(wave, date, Language_Questionnaire) %>%
  summarise(
```

```
    Sandu_mean = mean(Sandu, na.rm = TRUE),
    Sandu_sd = sd(Sandu, na.rm = TRUE),
    Sandu_n = sum(!is.na(Sandu)),
    Sandu_se = Sandu_sd / sqrt(Sandu_n),
    Sandu_lower = Sandu_mean - 1.96 * Sandu_se,
    Sandu_upper = Sandu_mean + 1.96 * Sandu_se,
    Dodon_mean = mean(Dodon, na.rm = TRUE),
    Dodon_sd = sd(Dodon, na.rm = TRUE),
    Dodon_n = sum(!is.na(Dodon)),
    Dodon_se = Dodon_sd / sqrt(Dodon_n),
    Dodon_lower = Dodon_mean - 1.96 * Dodon_se,
    Dodon_upper = Dodon_mean + 1.96 * Dodon_se,
    Sor_mean = mean(Sor, na.rm = TRUE),
    Sor_sd = sd(Sor, na.rm = TRUE),
    Sor_n = sum(!is.na(Sor)),
    Sor_se = Sor_sd / sqrt(Sor_n),
    Sor_lower = Sor_mean - 1.96 * Sor_se,
    Sor_upper = Sor_mean + 1.96 * Sor_se
  )

domestic_bop_language <- remove_missing(domestic_bop_language)
```

```
domestic_bop_language_long <- domestic_bop_language %>%
  pivot_longer(
    cols = c(Sandu_mean, Dodon_mean, Sor_mean, Sandu_lower, Sandu_upper, Dodon_lower, Dodon_upper, Sor_
    names_to = c("politician", ".value"),
    names_pattern = "(Sandu|Dodon|Sor)_(.*)"
  )
```

```
ggplot(domestic_bop_language_long, aes(x = date, y = mean,
                                       color = politician, group = politician)) +
  geom_ribbon(aes(ymin = lower, ymax = upper, fill = politician),
              alpha = 0.15, color = NA) +
  geom_line(linewidth = 0.8) +
  scale_y_reverse(limits = c(1,4)) +
  geom_point(size = 2) +
  facet_wrap(~Language_Questionnaire) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Models

```
Sandu_models <- list(
  "Age" = glm(Sandu ~ Age, data = bop_panel),
  "Gender" = glm(Sandu ~ Gender, data = bop_panel),
  "Ethnicity" = glm(Sandu ~ Ethnicity, data = bop_panel),
  "Language" = glm(Sandu ~ Language_Questionnaire, data = bop_panel),
  "Ecology" = glm(Sandu ~ Ecology, data = bop_panel),
  "Education" = glm(Sandu ~ Education, data = bop_panel),
  "Socioeconomic" = glm(Sandu ~ Socioeconomic, data = bop_panel),
  "Macro Region" = glm(Sandu ~ Macro_Region, data = bop_panel),
  "Good direction" = glm(Sandu ~ good, data = bop_panel),
  "War" = glm(Sandu ~ war, data = bop_panel),
  "Corruption" = glm(Sandu ~ corruption, data = bop_panel),
  "Interethnic" = glm(Sandu ~ interethnic, data = bop_panel),
  "Poverty" = glm(Sandu ~ poverty, data = bop_panel),
  "Year" = glm(Sandu ~ year, data = bop_panel),
  "Unification" = glm(Sandu ~ Unification_For, data = bop_panel),
  "All" = glm(Sandu ~ Age + Gender + Ethnicity + Language_Questionnaire + Ecology + Education + Socioeco
)

modelsummary(Sandu_models, stars = TRUE, coef_map = coef_model)
```

	Age	Gender	Ethnicity	Language	Ecology	Education	Soc
Age (per year)	0.000						
	(0.001)						
Woman		0.073***					
		(0.021)					
Romanian			−0.973***				
			(0.046)				
Russian			0.505***				
			(0.044)				
Ukrainian			0.474***				
			(0.037)				
Gagauz			0.718***				
			(0.050)				
Bulgarian			0.435***				
			(0.060)				
Russian questionnaire				0.503***			
				(0.021)			
Rural					−0.015		
					(0.022)		
Education: low						0.153***	
						(0.031)	
Education: high						−0.392***	
						(0.023)	
Socioeconomic: low							0
Socioeconomic: high							−
Centre							
North							
South							
Gagauzia							
Good (Direction)							
Worry: War							
Worry: Corruption							
Worry: Interethnic relations							
Worry: Poverty							

```r
Sor_models <- list(
  "Age" = glm(Sor ~ Age, data = bop_panel),
  "Gender" = glm(Sor ~ Gender, data = bop_panel),
  "Ethnicity" = glm(Sor ~ Ethnicity, data = bop_panel),
  "Language" = glm(Sor ~ Language_Questionnaire, data = bop_panel),
  "Ecology" = glm(Sor ~ Ecology, data = bop_panel),
  "Education" = glm(Sor ~ Education, data = bop_panel),
  "Socioeconomic" = glm(Sor ~ Socioeconomic, data = bop_panel),
  "Macro Region" = glm(Sor ~ Macro_Region, data = bop_panel),
  "Good direction" = glm(Sor ~ good, data = bop_panel),
  "Media (1)" = glm(Sor ~ top_TV, data = bop_panel),
  "Media (2)" = glm(Sor ~ top_internet, data = bop_panel),
  "Worries (1)" = glm(Sor ~ war, data = bop_panel),
  "Worries (2)" = glm(Sor ~ corruption, data = bop_panel),
  "Worries (3)" = glm(Sor ~ interethnic, data = bop_panel),
  "Worries (4)" = glm(Sor ~ poverty, data = bop_panel),
  "International politics (1)" = glm(Sor ~ Unification_For, data = bop_panel),
  "International politics (2)" = glm(Sor ~ Putin, data = bop_panel),
  "Year" = glm(Sor ~ year, data = bop_panel),
  "All" = glm(Sor ~ Age + Gender + Ethnicity + Language_Questionnaire + Ecology + Education + Socioecono
)

modelsummary(Sor_models, stars = TRUE, coef_map = coef_model)
```

```r
Dodon_models <- list(
  "Age" = glm(Dodon ~ Age, data = bop_panel),
  "Gender" = glm(Dodon ~ Gender, data = bop_panel),
  "Ethnicity" = glm(Dodon ~ Ethnicity, data = bop_panel),
  "Language" = glm(Dodon ~ Language_Questionnaire, data = bop_panel),
  "Ecology" = glm(Dodon ~ Ecology, data = bop_panel),
  "Education" = glm(Dodon ~ Education, data = bop_panel),
  "Socioeconomic" = glm(Dodon ~ Socioeconomic, data = bop_panel),
  "Macro Region" = glm(Dodon ~ Macro_Region, data = bop_panel),
  "Good direction" = glm(Dodon ~ good, data = bop_panel),
  "Media (1)" = glm(Dodon ~ top_TV, data = bop_panel),
  "Media (2)" = glm(Dodon ~ top_internet, data = bop_panel),
  "War" = glm(Dodon ~ war, data = bop_panel),
  "Corruption" = glm(Dodon ~ corruption, data = bop_panel),
  "Interethnic" = glm(Dodon ~ interethnic, data = bop_panel),
  "Poverty" = glm(Dodon ~ poverty, data = bop_panel),
  "Unification" = glm(Dodon ~ Unification_For, data = bop_panel),
  "Year" = glm(Dodon ~ year, data = bop_panel),
  "All" = glm(Dodon ~ Age + Gender + Ethnicity + Language_Questionnaire + Ecology + Education + Socioec
)

modelsummary(Dodon_models, stars = TRUE, coef_map = coef_model)
```

	Age	Gender	Ethnicity	Language	Ecology	Education	Soc
Age (per year)	−0.002***						
	(0.001)						
Woman		−0.058**					
		(0.020)					
Romanian			0.246***				
			(0.046)				
Russian			−0.372***				
			(0.043)				
Ukrainian			−0.402***				
			(0.036)				
Gagauz			−0.259***				
			(0.048)				
Bulgarian			−0.400***				
			(0.057)				
Russian questionnaire				−0.210***			
				(0.021)			
Rural					−0.085***		
					(0.020)		
Education: low						−0.087**	
						(0.033)	
Education: high						0.146***	
						(0.022)	
Socioeconomic: low							−
Socioeconomic: high							0
Centre							
North							
South							
Gagauzia							
Good (Direction)							
Worry: War							
Worry: Corruption							
Worry: Interethnic relations							
Worry: Poverty							

	Age	Gender	Ethnicity	Language	Ecology	Education	Soc
Age (per year)	−0.009***						
	(0.001)						
Woman		−0.017					
		(0.021)					
Romanian			0.502***				
			(0.044)				
Russian			−0.738***				
			(0.042)				
Ukrainian			−0.699***				
			(0.036)				
Gagauz			−0.912***				
			(0.047)				
Bulgarian			−0.647***				
			(0.058)				
Russian questionnaire				−0.294***			
				(0.021)			
Rural					−0.040+		
					(0.022)		
Education: low						−0.107***	
						(0.030)	
Education: high						0.299***	
						(0.022)	
Socioeconomic: low							−
Socioeconomic: high							0
Centre							
North							
South							
Gagauzia							
Good (Direction)							
Worry: War							
Worry: Corruption							
Worry: Interethnic relations							
Worry: Poverty							

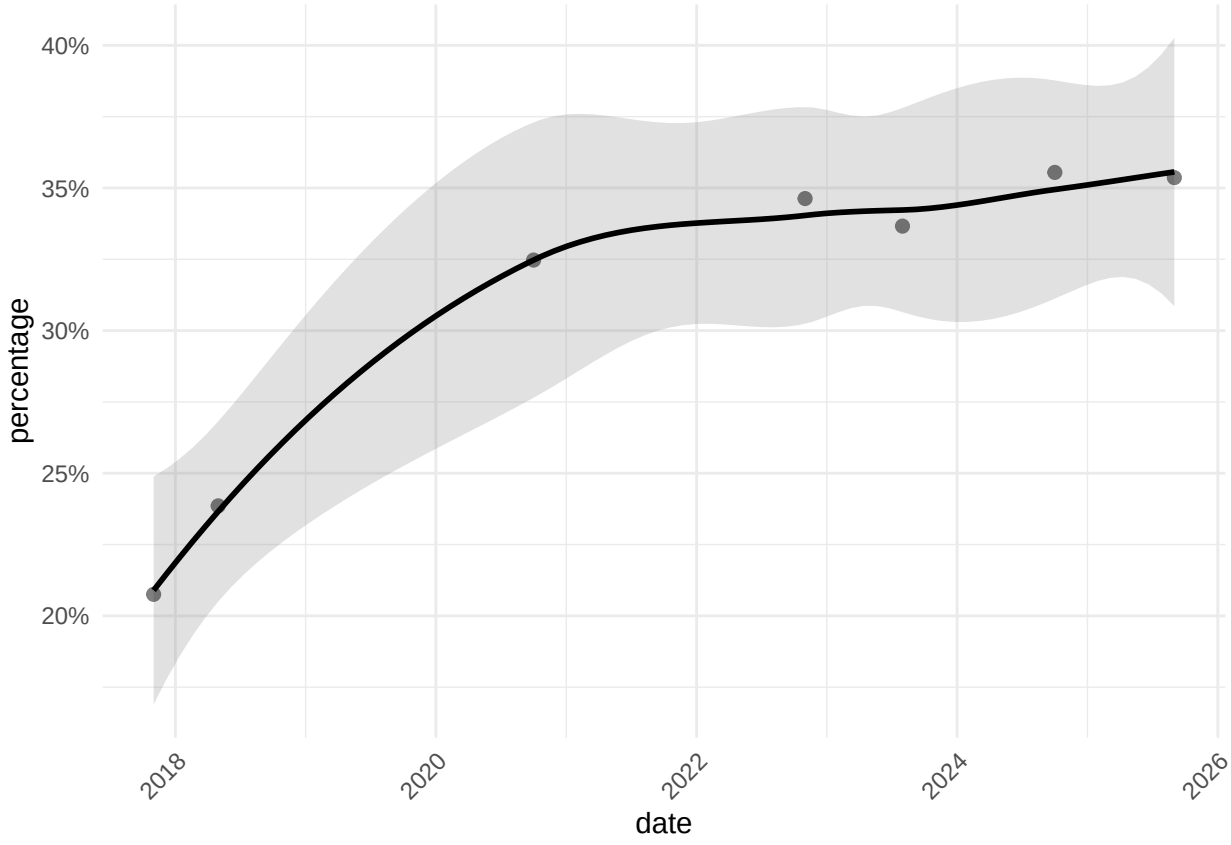
International politics

Unification

```
unification_support <- bop_panel %>%
  group_by(wave, date, Unification_For) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(percentage = n/sum(n))

unification_support <- remove_missing(unification_support)
```

```
unification_support %>%
  filter(Unification_For == 1) %>%
  ggplot(aes(x = date, y = percentage)) +
  geom_smooth(se = TRUE, color = "black", alpha = 0.3) +
  geom_point(size = 2, color = "black", alpha = 0.5) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
Unification_models <- list(
  "Age" = glm(Unification_For ~ Age, data = bop_panel, family = "binomial"),
```

```
  "Gender" = glm(Unification_For ~ Gender, data = bop_panel, family = "binomial"),
  "Ethnicity" = glm(Unification_For ~ Ethnicity, data = bop_panel, family = "binomial"),
  "Language" = glm(Unification_For ~ Language_Questionnaire, data = bop_panel, family = "binomial"),
  "Ecology" = glm(Unification_For ~ Ecology, data = bop_panel, family = "binomial"),
  "Education" = glm(Unification_For ~ Education, data = bop_panel, family = "binomial"),
  "Socioeconomic" = glm(Unification_For ~ Socioeconomic, data = bop_panel, family = "binomial"),
  "Macro Region" = glm(Unification_For ~ Macro_Region, data = bop_panel, family = "binomial"),
  "Year" = glm(Unification_For ~ year, data = bop_panel, family = "binomial"),
  "Nato" = glm(Unification_For ~ Nato_For, data = bop_panel, family = "binomial"),
  "EU" = glm(Unification_For ~ EU_For, data = bop_panel, family = "binomial"),
  "All" = glm(Unification_For ~ Age + Gender + Ethnicity + Language_Questionnaire + Ecology + Education
)

modelsummary(Unification_models, stars = TRUE)
```

EU support

```
# multilateral vs unilateral support
EU_support <- bop_panel %>%
  group_by(wave, date, EU_For) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(percentage = n/sum(n)) %>%
  filter(EU_For == 1)

EU_support$Variable <- "EU support (not binary)"

EU_support$EU_For <- NULL

EU_support2 <- bop_panel %>%
  group_by(wave, date, EU_VS) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(percentage = n/sum(n)) %>%
  filter(EU_VS == 1)

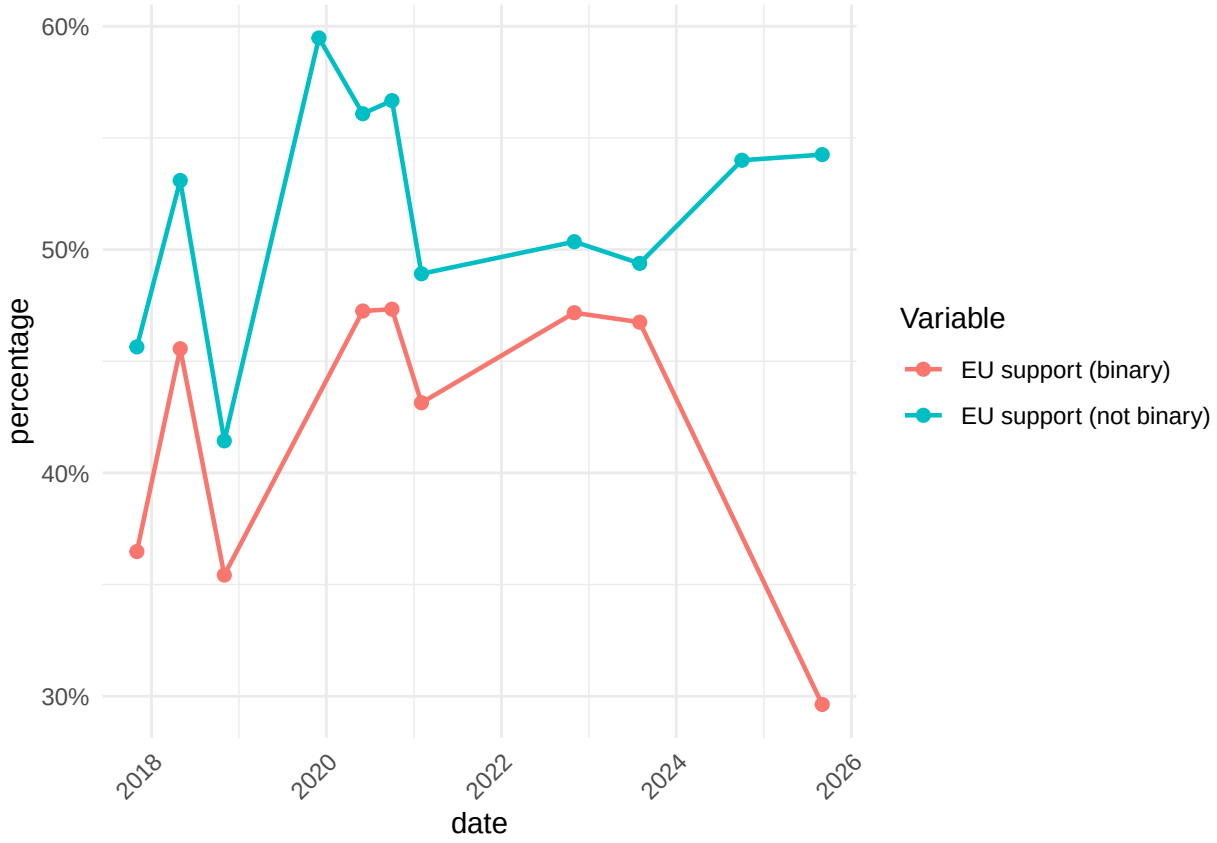
EU_support2$EU_VS <- NULL

EU_support2$Variable <- "EU support (binary)"

EU_support <- rbind(EU_support, EU_support2)
EU_support$IO <- "EU"
```

```
ggplot(EU_support, aes(x = date, y = percentage, color = Variable)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

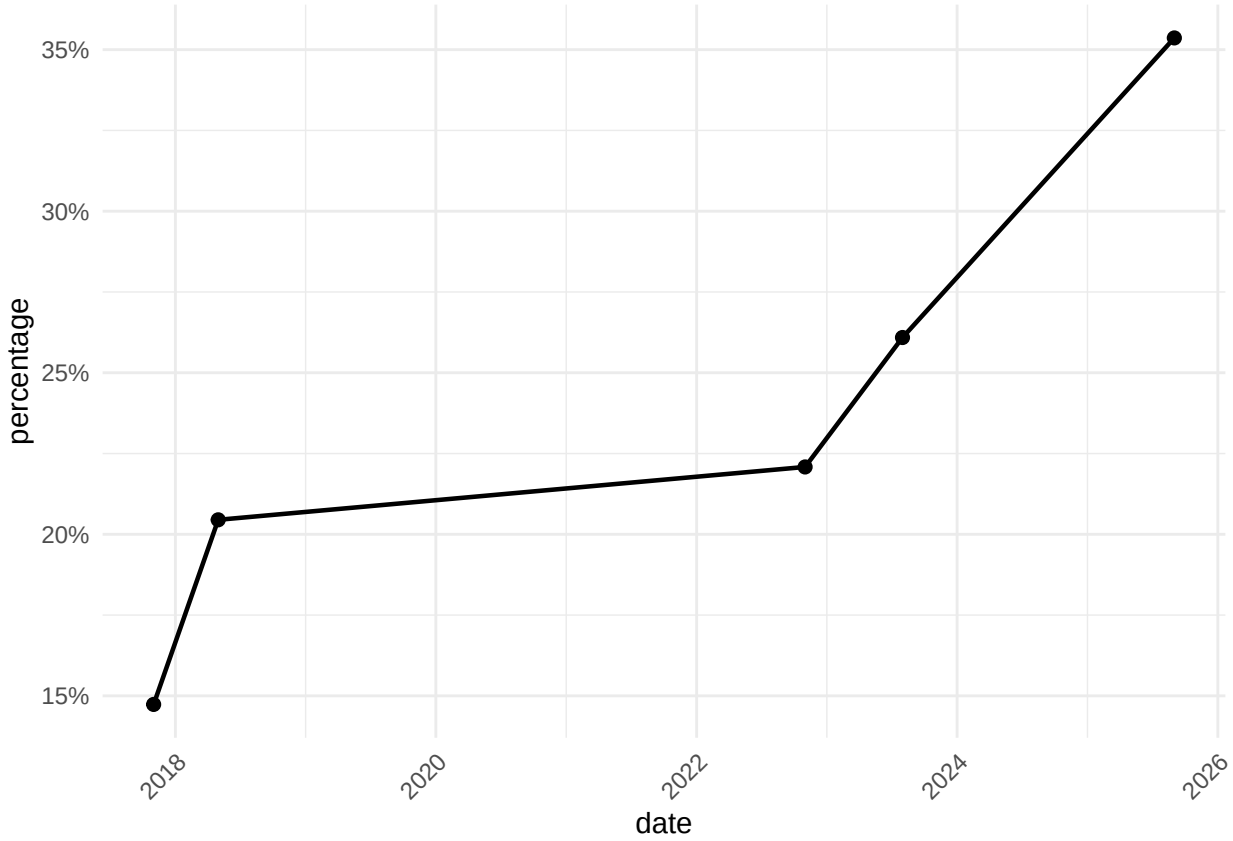
	Age	Gender	Ethnicity	Language	Ecology	Education	Socioeco
(Intercept)	−0.617***	−0.733***	−0.695***	−0.495***	−0.860***	−0.973***	−0.81
	(0.074)	(0.037)	(0.027)	(0.031)	(0.039)	(0.035)	(0.0
Age	−0.004**						
	(0.001)						
GenderWoman		−0.124*					
		(0.049)					
EthnicityRomanian			2.623***				
			(0.154)				
EthnicityRussian			−2.244***				
			(0.243)				
EthnicityUkrainian			−1.834***				
			(0.165)				
EthnicityGagauz			−3.953***				
			(0.581)				
EthnicityOther			−0.714*				
			(0.337)				
EthnicityBulgarian			−2.027***				
			(0.328)				
EthnicityDK/NR			−0.558				
			(0.802)				
Language_QuestionnaireRussian				−0.789***			
				(0.052)			
Ecologyrural					0.040		
					(0.053)		
Educationlow						−0.101	
						(0.074)	
Educationhigh						0.531***	
						(0.053)	
Socioeconomiclow							−0.25
							(0.0
Socioeconomichigh							0.311
							(0.0
Macro_RegionCentre							
Macro_RegionNorth							
Macro_RegionSouth							
Macro_RegionGagauzia							
year							
Nato_For							



NATO support

```
NATO_support <- bop_panel %>%
  group_by(wave, date, Nato_For) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(percentage = n/sum(n)) %>%
  filter(Nato_For == 1)
```

```
ggplot(NATO_support, aes(x = date, y = percentage)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Eurasia support

```
Eurasia_support <- bop_panel %>%
  group_by(wave, date, Eurasia_For) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(percentage = n/sum(n)) %>%
  filter(Eurasia_For == 1)

Eurasia_support$Variable <- "Eurasia support (not binary)"

Eurasia_support$Eurasia_For <- NULL

Eurasia_support2 <- bop_panel %>%
  group_by(wave, date, Eurasia_VS) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(wave, date) %>%
  mutate(percentage = n/sum(n)) %>%
  filter(Eurasia_VS == 1)

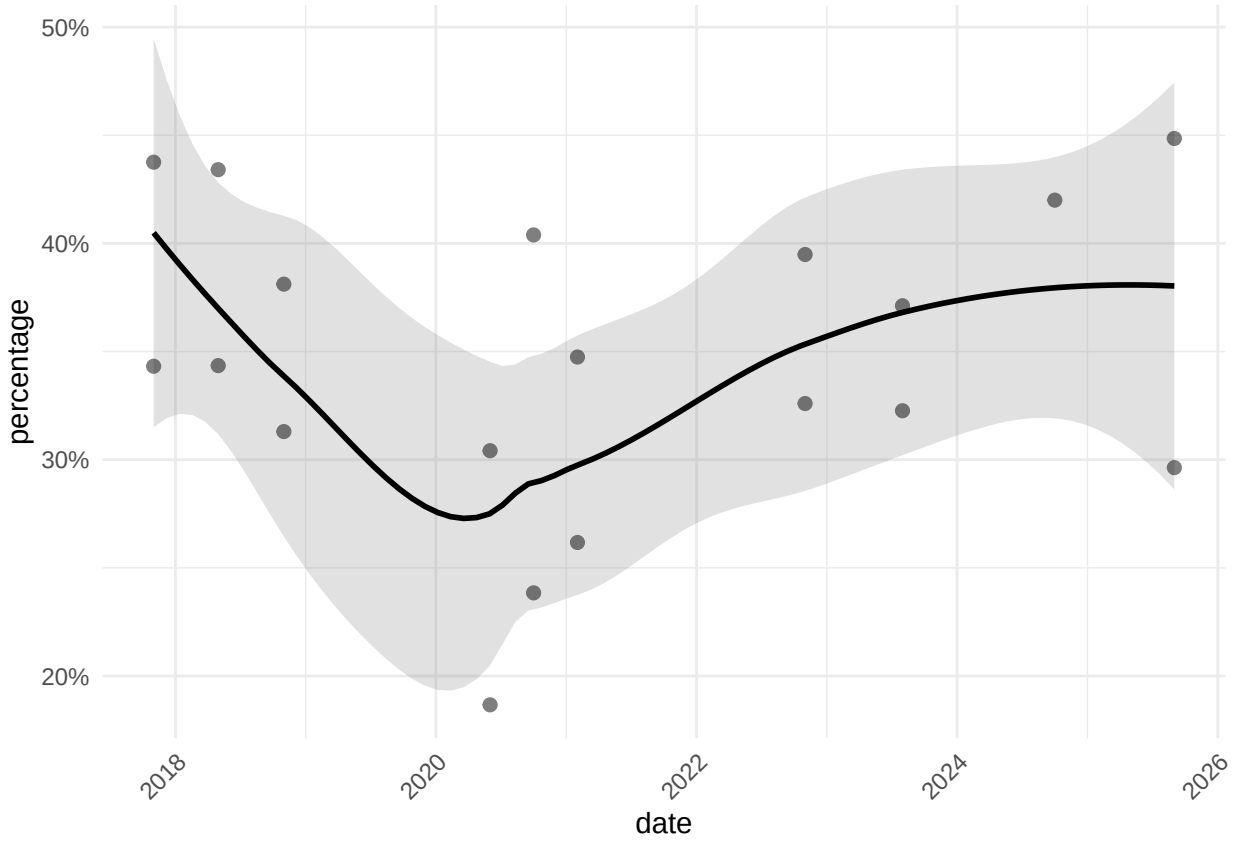
Eurasia_support2$Eurasia_VS <- NULL

Eurasia_support2$Variable <- "Eurasia support (binary)"
```

```
Eurasia_support <- rbind(Eurasia_support, Eurasia_support2)

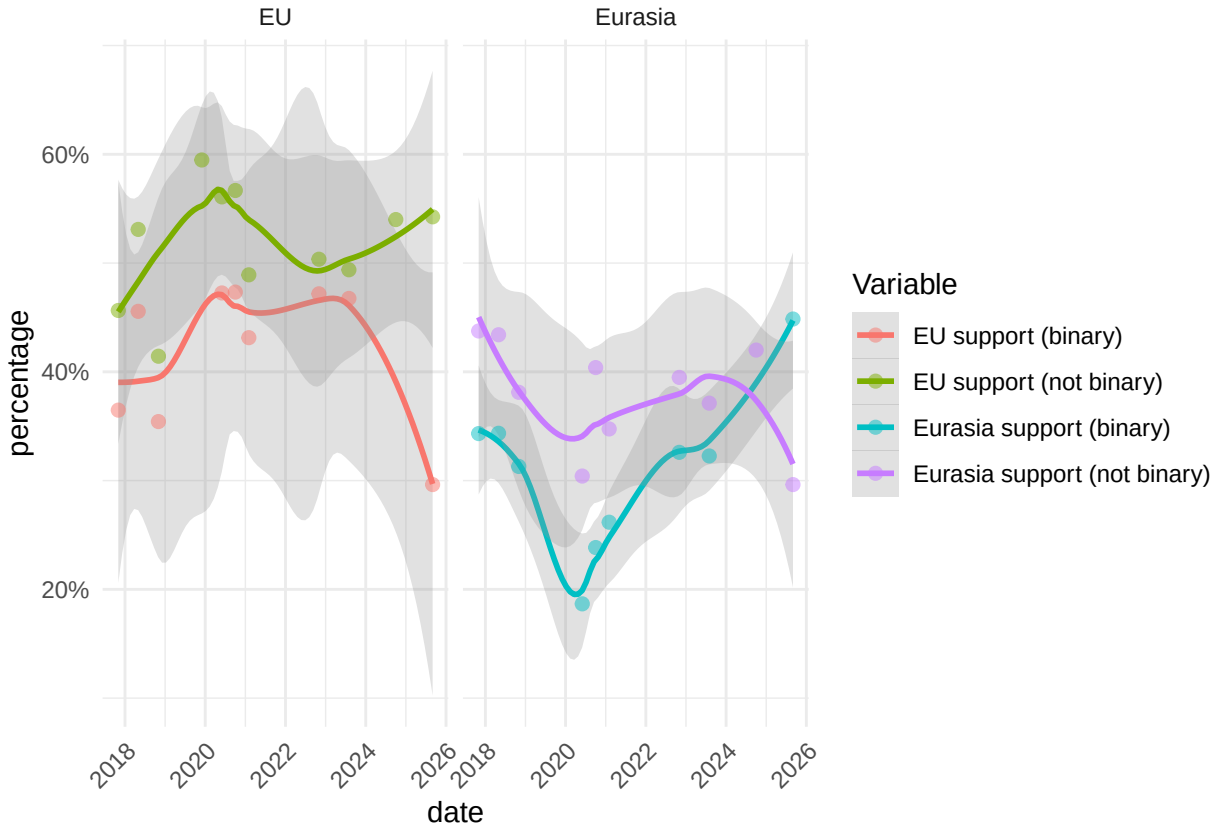
Eurasia_support$IO <- "Eurasia"
```

```
ggplot(Eurasia_support, aes(x = date, y = percentage, color = Variable)) +
  geom_smooth(se = TRUE, color = "black", alpha = 0.3) +
  geom_point(size = 2, color = "black", alpha = 0.5) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
IO_support <- rbind(EU_support, Eurasia_support)

ggplot(IO_support, aes(x = date, y = percentage, color = Variable)) +
  geom_smooth(se = TRUE, alpha = 0.3) +
  geom_point(size = 2, alpha = 0.5) +
  facet_wrap(~IO) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



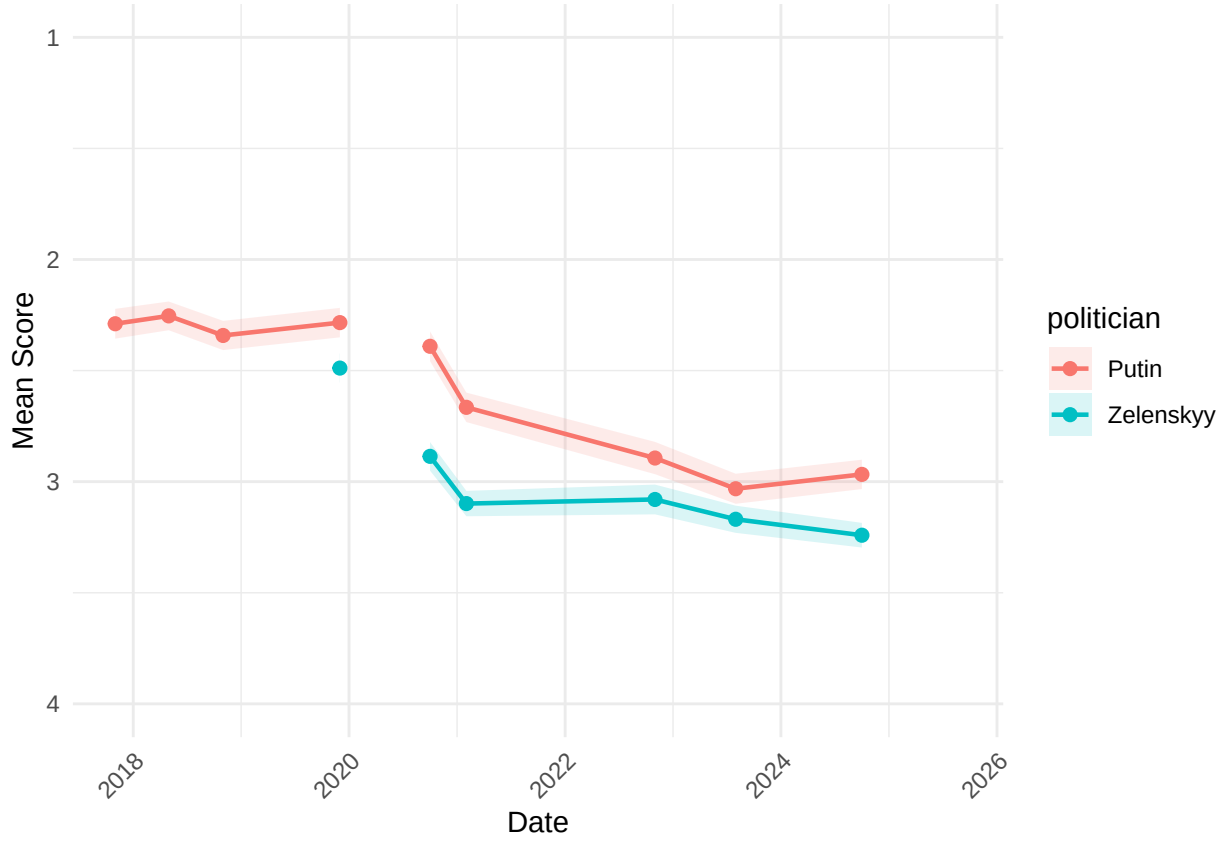
Putin & Zelenskyy

```
international_bop <- bop_panel %>%
  group_by(wave, date) %>%
  summarise(
    Putin_mean = mean(Putin, na.rm = TRUE),
    Putin_sd = sd(Putin, na.rm = TRUE),
    Putin_n = sum(!is.na(Putin)),
    Putin_se = Putin_sd / sqrt(Putin_n),
    Putin_lower = Putin_mean - 1.96 * Putin_se,
    Putin_upper = Putin_mean + 1.96 * Putin_se,
    Zelenskyy_mean = mean(Zelenskyy, na.rm = TRUE),
    Zelenskyy_sd = sd(Zelenskyy, na.rm = TRUE),
    Zelenskyy_n = sum(!is.na(Zelenskyy)),
    Zelenskyy_se = Zelenskyy_sd / sqrt(Zelenskyy_n),
    Zelenskyy_lower = Zelenskyy_mean - 1.96 * Zelenskyy_se,
    Zelenskyy_upper = Zelenskyy_mean + 1.96 * Zelenskyy_se
  )
```

```
international_bop <- international_bop %>%
  pivot_longer(
    cols = c(Putin_mean, Putin_sd, Putin_n, Putin_se, Putin_lower, Putin_upper,
             Zelenskyy_mean, Zelenskyy_sd, Zelenskyy_n, Zelenskyy_se, Zelenskyy_lower, Zelenskyy_upper)
    names_to = c("politician", ".value"),
```

```
    names_pattern = "(Putin|Zelenskyy)_(.*)"
  )
```

```
ggplot(international_bop, aes(x = date, y = mean, color = politician,
                              group = politician)) +
  geom_ribbon(aes(ymin = lower, ymax = upper, fill = politician),
              alpha = 0.15, color = NA) +
  geom_line(linewidth = 0.8) +
  scale_y_reverse(limits = c(1,4)) +
  geom_point(size = 2) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "Date",
       y = "Mean Score")
```




```
Putin_models <- list(
  "Age" = glm(Putin ~ Age, data = bop_panel),
  "Gender" = glm(Putin ~ Gender, data = bop_panel),
  "Ethnicity" = glm(Dodon ~ Ethnicity, data = bop_panel),
  "Language" = glm(Putin ~ Language_Questionnaire, data = bop_panel),
  "Ecology" = glm(Putin ~ Ecology, data = bop_panel),
  "Education" = glm(Putin ~ Education, data = bop_panel),
  "Socioeconomic" = glm(Putin ~ Socioeconomic, data = bop_panel),
  "Macro Region" = glm(Putin ~ Macro_Region, data = bop_panel),
  "Bad direction" = glm(Putin ~ bad, data = bop_panel),
```

```
  "Media (1)" = glm(Putin ~ top_TV, data = bop_panel),
  "Media (2)" = glm(Putin ~ top_internet, data = bop_panel),
  "War" = glm(Putin ~ war, data = bop_panel),
  "Corruption" = glm(Putin ~ corruption, data = bop_panel),
  "Interethnic" = glm(Putin ~ interethnic, data = bop_panel),
  "Poverty" = glm(Putin ~ poverty, data = bop_panel),
  "Unification" = glm(Putin ~ Unification_For, data = bop_panel),
  "Year" = glm(Putin ~ year, data = bop_panel),
  "All" = glm(Putin ~ Age + Gender + Ethnicity + Language_Questionnaire + Ecology + Education + Socioeco
)

modelsummary(Putin_models, stars = TRUE, coef_map = coef_model)
```

```
Zelenskyy_models <- list(
  "Age" = glm(Zelenskyy ~ Age, data = bop_panel),
  "Gender" = glm(Zelenskyy ~ Gender, data = bop_panel),
  "Ethnicity" = glm(Zelenskyy ~ Ethnicity, data = bop_panel),
  "Language" = glm(Zelenskyy ~ Language_Questionnaire, data = bop_panel),
  "Ecology" = glm(Zelenskyy ~ Ecology, data = bop_panel),
  "Education" = glm(Zelenskyy ~ Education, data = bop_panel),
  "Socioeconomic" = glm(Zelenskyy ~ Socioeconomic, data = bop_panel),
  "Macro Region" = glm(Zelenskyy ~ Macro_Region, data = bop_panel),
  "Bad direction" = glm(Zelenskyy ~ bad, data = bop_panel),
  "Media (1)" = glm(Zelenskyy ~ top_TV, data = bop_panel),
  "Media (2)" = glm(Zelenskyy ~ top_internet, data = bop_panel),
  "War" = glm(Zelenskyy ~ war, data = bop_panel),
  "Corruption" = glm(Zelenskyy ~ corruption, data = bop_panel),
  "Interethnic" = glm(Zelenskyy ~ interethnic, data = bop_panel),
  "Poverty" = glm(Zelenskyy ~ poverty, data = bop_panel),
  "Unification" = glm(Zelenskyy ~ Unification_For, data = bop_panel),
  "Year" = glm(Zelenskyy ~ year, data = bop_panel),
  "All" = glm(Zelenskyy ~ Age + Gender + Ethnicity + Language_Questionnaire + Ecology + Education + Soci
)

modelsummary(Zelenskyy_models, stars = TRUE, coef_map = coef_model)
```

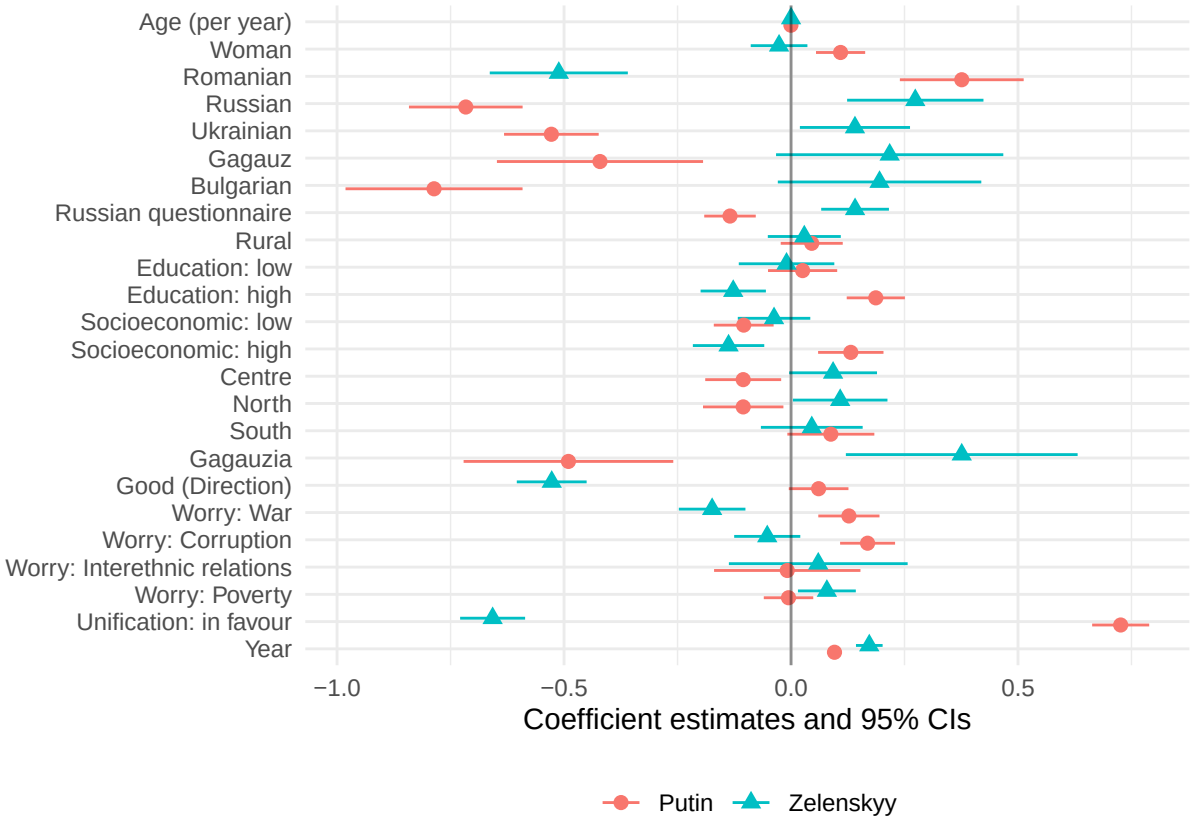
```
international_leaders <- list(
  "Putin" = glm(Putin ~ Age + Gender + Ethnicity + Language_Questionnaire +
                  Ecology + Education + Socioeconomic + Macro_Region +
                  Unification_For + good + poverty + corruption +
                  interethnic + war + year,
                data = bop_panel),
  "Zelenskyy" = glm(Zelenskyy ~ Age + Gender + Ethnicity + Language_Questionnaire +
                      Ecology + Education + Socioeconomic + Macro_Region +
                      Unification_For + good + poverty + corruption +
                      interethnic + war + year,
                data = bop_panel)
)


modelplot(international_leaders, coef_map = coef_plot) +
  aes(shape = model) +
```

	Age	Gender	Ethnicity	Language	Ecology	Education	Soc
Age (per year)	−0.005***						
	(0.001)						
Woman		−0.010					
		(0.024)					
Romanian			0.502***				
			(0.044)				
Russian			−0.738***				
			(0.042)				
Ukrainian			−0.699***				
			(0.036)				
Gagauz			−0.912***				
			(0.047)				
Bulgarian			−0.647***				
			(0.058)				
Russian questionnaire				−0.364***			
				(0.024)			
Rural					−0.077**		
					(0.025)		
Education: low						−0.107**	
						(0.034)	
Education: high						0.414***	
						(0.026)	
Socioeconomic: low							−
Socioeconomic: high							0
Centre							
North							
South							
Gagauzia							
Bad (Direction)							
Worry: War							
Worry: Corruption							
Worry: Interethnic relations							
Worry: Poverty							

	Age	Gender	Ethnicity	Language	Ecology	Education	Socioeconom
Age (per year)	0.003***						
	(0.001)						
Woman		0.021					
		(0.027)					
Romanian			−0.943***				
			(0.056)				
Russian			0.464***				
			(0.063)				
Ukrainian			0.371***				
			(0.049)				
Gagauz			0.712***				
			(0.059)				
Bulgarian			0.687***				
			(0.095)				
Russian questionnaire				0.403***			
				(0.028)			
Rural					0.088**		
					(0.030)		
Education: low						0.023	
						(0.045)	
Education: high						−0.266***	
						(0.028)	
Socioeconomic: low							0.118**
							(0.037)
Socioeconomic: high							−0.262***
							(0.035)
Centre							
North							
South							
Gagauzia							
Bad (Direction)							
Worry: War							
Worry: Corruption							
Worry: Interethnic relations							
Worry: Poverty							

```
  geom_vline(xintercept = 0, color = 'black', alpha = 0.4) +
  theme(legend.position = "bottom") +
  xlab("Coefficient estimates and 95% CIs")
```



PCA / factor analysis ?